Genome-Scale Context, Stoichiometric Refinement, and Neuro-Symbolic Reasoning for Proteome Function Annotation

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Abstract

Per-protein function predictors ignore genomic context, leaving fragmented pathways, isolated complex subunits, and clade-inappropriate Gene Ontology (GO) assignments, whereas negative-cost mixed-integer linear programming (MILP) gap-fillers add hundreds of zero-flux reactions without gene-locus attribution. We present Genome-Scale Protein Annotation (GSPA v2.1.0), a four-stage framework coupling multi-channel probability calibration (Stage 0), whole-proteome Bayesian log-odds priors (Stage 1), semantic ancestor relaxation of taxonomic and Uberon/PO/CL anatomical constraints (Stage 2A), positive-cost parsimonious flux balance analysis (pFBA) with locus responsibility against an unassigned-gene hypothesis (Stage 2B), and selective guarded language-model disambiguation on $< 2.5\%$ of proteins (Stage 3). Across eight benchmark suites, Stage 0 establishes bulk protein-centric accuracy while Stages 1–3 eliminate taxonomic/anatomical violations (1,009 endogenous and 1,452 stress-test eukaryotic violations), improve eukaryotic $F_{\max}$ (+0.0021 endogenous CAFA $F_{\max}$ mean; +0.0006 to +0.0157 transfer-evaluated micro $F_{\max}$ over naive deletion across multicellular animals and vascular plants under unmodified Swiss-Prot ground truth), certify 100% of pFBA reaction additions (5.70 reactions/genome in 18.01 s/genome full certification vs. 215.02 (68.62%) uncertified zero-flux additions/genome under negative-cost MILP) across 108 genomes, improve `Price-149` enzyme Top-1 hits under both single-predictor refinement (CLEAN 60 → 63, MAPred 50 → 56, DeepProZyme 49 → 51 with cognate SLR, GraphEC 18 → 20) and multi-predictor refinement (+0.0518 to +0.1987 $F_{\max}$), match gapseq binary `BacDive` accuracy while lifting blind carbon ROC-AUC to 0.8567, Track 2 phenotype-anchored F_1 to 0.8871 (carbon) and 0.8545 (enzyme), and genus-blocked hybrid F_1 to 0.8710 (carbon) and 0.8713 (enzyme), and lift ambiguous leave-reaction-out Hit@1 from 34.0% to 96.7% (3-seed mean).

1 Introduction

Automated annotation of protein function from genome assemblies is a central problem in computational biology, underpinning comparative genomics, genome-scale metabolic modeling, and systems biology[1,2]. Two major methodological paradigms have been developed for assigning functional terms from the Gene Ontology (GO)[3] and Enzyme Commission (EC) nomenclature. Production genome annotation pipelines—such as the NCBI Prokaryotic Genome Annotation Pipeline (PGAP)[4,5], Bakta[6], Prokka[7], eggNOG-mapper[8], Pharokka[9], and InterProScan[10,11]—rely on sequence alignment[12], profile hidden Markov models (HMMs)[13], and curated orthology rules to assign annotations protein by protein. Concurrently, deep learning and protein language model (PLM) predictors[14–16] evaluated in the Critical Assessment of Functional Annotation (CAFA) challenges[17–19]—including DeepGOPlus[20], DeepFRI[21], metagenomic-deepFRI[22], DeepGOZero[23], DeepGO-SE[24], ProteInfer[25], SPROF-GO[26], CLEAN[27], DeepProZyme[28], EnzBERT[29], MAPred[30], GraphEC[31], and TopEC[32]—predict calibrated GO or EC probabilities from individual amino acid sequences or predicted three-dimensional structures[33].

Despite their differences in representation, both paradigms share a fundamental modeling assumption: they predict the function of each protein p_i in a proteome $\mathcal{P} = \{p_1, \ldots, p_N\}$ conditionally independently of the remaining proteins $\mathcal{P} \setminus \{p_i\}$ and of the organism's broader biological architecture. Evaluated at the scale of a whole genome, this independence assumption produces five systematic failure modes: (1) essential cellular machinery required by viable organisms in a domain of life (such as translation, tRNA aminoacylation, or DNA replication initiation) can remain unannotated when divergent orthologs fall just below strict per-protein homology thresholds; (2) predicted enzymatic repertoires leave stoichiometric holes in otherwise complete metabolic pathways[34,35] when individual enzymes reside in the sequence-identity twilight zone ($< 30\%$ identity); (3) obligate heteromeric protein complexes (such as ATP synthase, NADH dehydrogenase, urease, or photosystem reaction centers[36,37]) suffer from both missing subunits and false-

positive singleton subunit calls triggered by promiscuous domain hits; (4) sequence similarity and PLM embedding transfer across distant homologs routinely transfer clade-inappropriate GO terms that violate formal taxonomic constraints[38]; and (5) in fungi, plants, and animals, homology transfer frequently assigns developmental or physiological biological processes—such as *heart development* (`GO:0007507`), *flower development* (`GO:0009908`), *neural crest cell migration* (`GO:0001755`), or *vascular transport* (`GO:0010232`)—to organisms that lack the corresponding anatomical structures.

Comparative genomics and systems biology offer complementary constraints that can resolve these failures. Gene neighborhood conservation, co-directional operon structure, and phylogenetic profiles capture functional coupling across prokaryotic pathways and complexes[39–43], as recently learned by genomic language models such as gLM[44], gLM2[45], Evo[46], and contextualized bacterial protein models[47,48]. In metabolic network reconstruction, tools such as Pathway Tools[49,50], ModelSEED[51,52], ProbAnno[53], CarveMe[54], gapseq[55], GapMind[56,57], Reconstructor[58], and CanOE[59] use stoichiometric network connectivity to fill reaction gaps. However, automated top-down reconstructions can still mispredict auxotrophies and catabolic phenotypes when gene–protein–reaction (GPR) associations are noisy[60]. Moreover, reaction-level MILP gap-fillers—including CarveMe[54] and our own prior reaction-indicator MILP formulation METEOR[61] (developed by overlapping authors of the present study)—optimize binary reaction indicators $y_j \in \{0,1\}$ without distinguishing isozymes from obligate heteromeric complex subunits, assign negative objective costs ($c_j^{\text{add}} < 0$) to high-scoring candidate reactions (which activates hundreds of zero-flux gap-fill additions), and apply uniform reaction-to-protein probability boosts that cannot disambiguate which specific gene locus encodes a missing enzymatic step[61]. In bio-ontology engineering, formal taxon constraints (`only_in_taxon` and `never_in_taxon`)[38] and cross-product logical definitions linking GO biological processes to the Uberon multi-species anatomy ontology[62–65], the Plant Ontology (PO)[66], and the Cell Ontology (CL)[67] provide a formal logical foundation for testing whether an organism possesses the anatomical structures entailed by a functional annotation. Concurrently, tool-using biological language agents such as GeneAgent[68], ProtAgents[69], SPIRES[70], GORetriever[71], and ProtST[72] have demonstrated strong capability in synthesizing multi-modal evidence, provided they are invoked selectively on ambiguous loci and gated by formal symbolic verification.

Here we present Genome-Scale Protein Annotation (GSPA v2.1.0), a unified four-stage framework for proteome-wide function prediction (Figure 1). Stage 0 calibrates and pools per-protein evidence channels via Noisy-OR (or ingests any single standalone GO or EC predictor). Stage 1 iteratively refines annotation log-odds across the proteome using six genome-scale priors: core essentiality, pathway and complex coherence, SAT-based taxonomic consistency, GO–Uberon/PO/CL anatomical constraints, metabolic gap-filling with reaction-local genomic context (RLGC), and operon neighborhood coupling. Stage 2 combines a deterministic symbolic enforcer (Stage 2A) that eliminates taxonomic and anatomical violations via semantic ancestor relaxation with a flux-certified stoichiometric locus refiner (Stage 2B) that couples sparse GPR aggregation, positive-cost lexicographic parsimonious FBA (pFBA)[73], and backward locus responsibility against an explicit unassigned-gene hypothesis $h_\varnothing$. Stage 3 provides an optional, selectively triggered neuro-symbolic agent using an open-weight instruction language model (`Qwen/Qwen3.8-27B-FP8`, served as `qwen3.8-27b`) gated by a deterministic symbolic ontology guardrail. We evaluate GSPA across eight benchmark suites (Tables 1–8; full provenance mapping in `results/PROVENANCE_MAP.md`).

2 Results

2.1 Multi-evidence integration and genome-scale prior ablation (Suites 1 and 2)

We first evaluated how Stage 0 multi-channel pooling and Stages 1–2 genome-scale priors affect protein-centric accuracy and structural genome coherence across nine bacterial and archaeal reference genomes (*Escherichia coli* K-12, *E. coli* O157:H7, *Bacillus subtilis* 168, *Pseudomonas aeruginosa* PAO1, *Mycobacterium tuberculosis* H37Rv, *Helicobacter pylori* 26695, *Mycoplasma genitalium* G37, *Synechocystis* sp. PCC 6803, and *Methanocaldococcus jannaschii* DSM 2661; 81 runs; Figure 2 and Table 1). For high-throughput multi-genome benchmarks (the 9-genome, 13-genome, 88-genome, and 1,000-genome panels), Stage 0 defaults to the 3 fast sequence/domain channels (DIAMOND, InterProScan, and Pfam HMMER; 5.4–12.7 s/genome), whereas on the nested 21-genome neural panel (which contains all 13 panel genomes plus 8 additional reference proteomes, and is itself nested within the 88-genome four-kingdom cohort) we evaluate both the 3-channel core and all 7 channels including ESM-2/ProtT5 kNN, ProteInfer, and FoldSeek+ProstT5 3Di. Because roughly 85% of annotated proteins in the 9 reference organisms carry high-confidence Swiss-Prot homology ($P_0 \geq 0.90$), Stage 0 Noisy-OR pooling (`A1_noisy_or`) establishes the bulk protein-centric baseline: micro $F_{\max} = 0.7631 \pm 0.0817$ (sample s.d., ddof = 1, across 9 genomes; 0.2853 ± 0.1038 against experimental-only truth) and CAFA $F_{\max} = 0.8041 \pm 0.0650$ (0.2363 ± 0.1489 experimental-only). Across the full Stage 1+2 pipeline (`A9_stage2_rel`), pair-level micro $F_{\max}$ increases slightly to 0.7660 ± 0.0791 (0.2847 ± 0.1016 experimental-only; two-sided exact paired Wilcoxon $p = 0.0391$) while CAFA $F_{\max}$ is 0.8037 ± 0.0647 (0.2366 ± 0.1494 experimental-only; two-sided exact paired Wilcoxon $p = 0.1289$, with slight

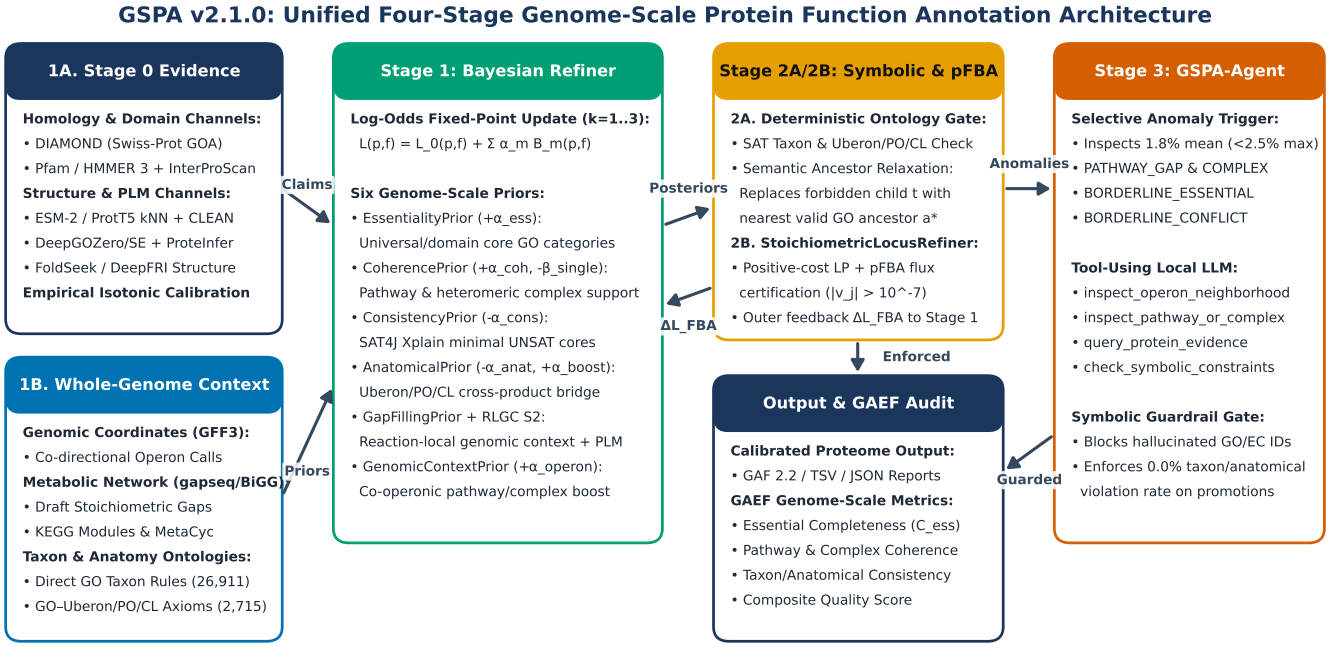


Figure 1. Unified four-stage neuro-symbolic architecture of GSPA v2.1.0. (**Stage 0**) Ingests calibrated per-protein evidence channels (sequence homology, domain HMMs, structure similarity, and protein language model embeddings) or any single standalone predictor alongside whole-genome context (GFF3 operon coordinates, draft metabolic network gaps, 26,911 direct GO taxon constraint rules [13,819 `only_in_taxon` and 13,092 `never_in_taxon`] across 18,159 GO terms, and 2,715 GO–Uberon/PO/CL anatomical axioms across 2,674 GO terms and 1,294 anatomical classes across 30 object properties linked to 8,331 taxon constraints across 1,247 anatomical classes). (**Stage 1**) The iterative Bayesian refiner updates log-odds posteriors across all protein–term pairs via six genome-scale priors and reaction-local genomic context (RLGC). (**Stage 2**) The deterministic symbolic enforcer (Stage 2A) resolves surviving taxonomic and anatomical contradictions through semantic ancestor relaxation and prunes isolated heteromeric complex singletons, while the stoichiometric locus refiner (Stage 2B) performs positive-cost lexicographic pFBA repair and backward locus attribution. (**Stage 3**) An optional selective agentic layer inspects the $< 2.5\%$ of loci with unresolved pathway gaps, incomplete complexes, or borderline conflicts using deterministic genomic tools and a symbolic ontology guardrail.

Table 1. Systematic ablation of GSPA Stage 1 priors and Stage 2 symbolic enforcement across nine reference genomes (Suite 2, 81 runs; source: `results/pillar1_2_ablation_summary.tsv` and `results/pillar1_2_perturbation_summary.tsv`). Values report means (± sample s.d., ddof = 1, in text) across *E. coli* K-12 (hyperparameter calibration genome), *E. coli* O157:H7, *B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *M. genitalium* G37, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661; the bottom row reports the 8-genome out-of-sample mean excluding *E. coli* K-12. Micro $F_{\max}$ and CAFA $F_{\max}$ report full GOA (`All`) and experimental-only (`Exp`) ground-truth evaluations. C_{cons} (Viol.) reports mean consistency score $\max(0, 1 - 0.01V)$ and total baseline violations across the panel; Perturb. Supp. reports suppression of the 125 post-calibration active violations (122/125 = 97.6%), with suppression across all 900 raw injected claims (897/900 = 99.7%) in parentheses.

Config	Active Priors / Enforcement	Micro $F_{\max}$ All (Exp)	CAFA $F_{\max}$ All (Exp)	C_{ess}	C_{path}	C_{cplx}	C_{cons} (Viol.)	GAEF Score	Perturb. Supp.
`A1_noisy_or`	Stage 0 Noisy-OR baseline	0.7631 (0.2853)	0.8041 (0.2363)	0.9400	0.4975	0.8636	0.9722 (25)	0.8459	0.0% (86.1%)
`A2_essentiality`	+ Essentiality prior ($\alpha_1 = 1.5$)	0.7632 (0.2853)	0.8040 (0.2363)	0.9444	0.4975	0.8636	0.9722 (25)	0.8472	0.0% (86.1%)
`A3_coherence`	+ Coherence prior ($\alpha_2 = 1.5$)	0.7629 (0.2824)	0.8025 (0.2354)	0.9400	0.5052	0.8636	0.9722 (25)	0.8474	0.0% (86.1%)
`A4_gap_filling`	+ Gap-filling prior ($\alpha_5 = 1.2$)	0.7631 (0.2853)	0.8038 (0.2361)	0.9400	0.4977	0.8636	0.9722 (25)	0.8459	0.0% (86.1%)
`A5_genomic_ctx`	+ Operon context prior ($\alpha_6 = 1.0$)	0.7630 (0.2851)	0.8039 (0.2363)	0.9400	0.4992	0.8636	0.9722 (25)	0.8462	0.0% (86.1%)
`A6_taxon_anat`	+ Consistency + Anatomical priors	**0.7681 (0.2894)**	**0.8071 (0.2389)**	0.9400	0.4973	0.8636	**1.0000 (0)**	0.8542	**97.6% (99.7%)**
`A7_stage1_all`	All six Stage 1 priors combined	0.7660 (0.2847)	0.8037 (0.2366)	**0.9479**	**0.5146**	**0.8852**	**1.0000 (0)**	**0.8643**	97.6% (99.7%)
`A8_stage2_rem`	Stage 1 + Stage 2A (remove)	0.7660 (0.2847)	0.8037 (0.2366)	**0.9479**	**0.5146**	**0.8852**	**1.0000 (0)**	**0.8643**	**97.6% (99.7%)**
`A9_stage2_rel`	Stage 1 + Stage 2A (relax)	0.7660 (0.2847)	0.8037 (0.2366)	**0.9479**	**0.5146**	**0.8852**	**1.0000 (0)**	**0.8643**	**97.6% (99.7%)**
8-Genome Non-*E. coli* Mean: `A1_noisy_or` → `A9_stage2_rel`		0.7718 → 0.7749	0.8143 → 0.8140	0.9325 → 0.9414	0.4834 → 0.5012	0.8743 → 0.8865	0.9738 (21) → 1.0000 (0)	0.8434 → 0.8599	0.0% → 97.6%

decreases of 0.0001–0.0016 in 8 of 9 genomes; Table 1). Excluding the hyperparameter development genome *E. coli* K-12 yields the same pattern across the 8 out-of-sample genomes: micro $F_{\max}$ rises from 0.7718 to 0.7749, CAFA $F_{\max}$ is 0.8143 vs. 0.8140 (0.2140 → 0.2143 experimental-only), C_{path} rises from 0.4834 to 0.5012, C_{cplx} rises from 0.8743 to 0.8865, and the internal GAEF structural score rises from 0.8434 to 0.8599.

Examining individual priors on the 9-genome panel (Table 1), the essentiality prior (`A2`) lifts mean core functional

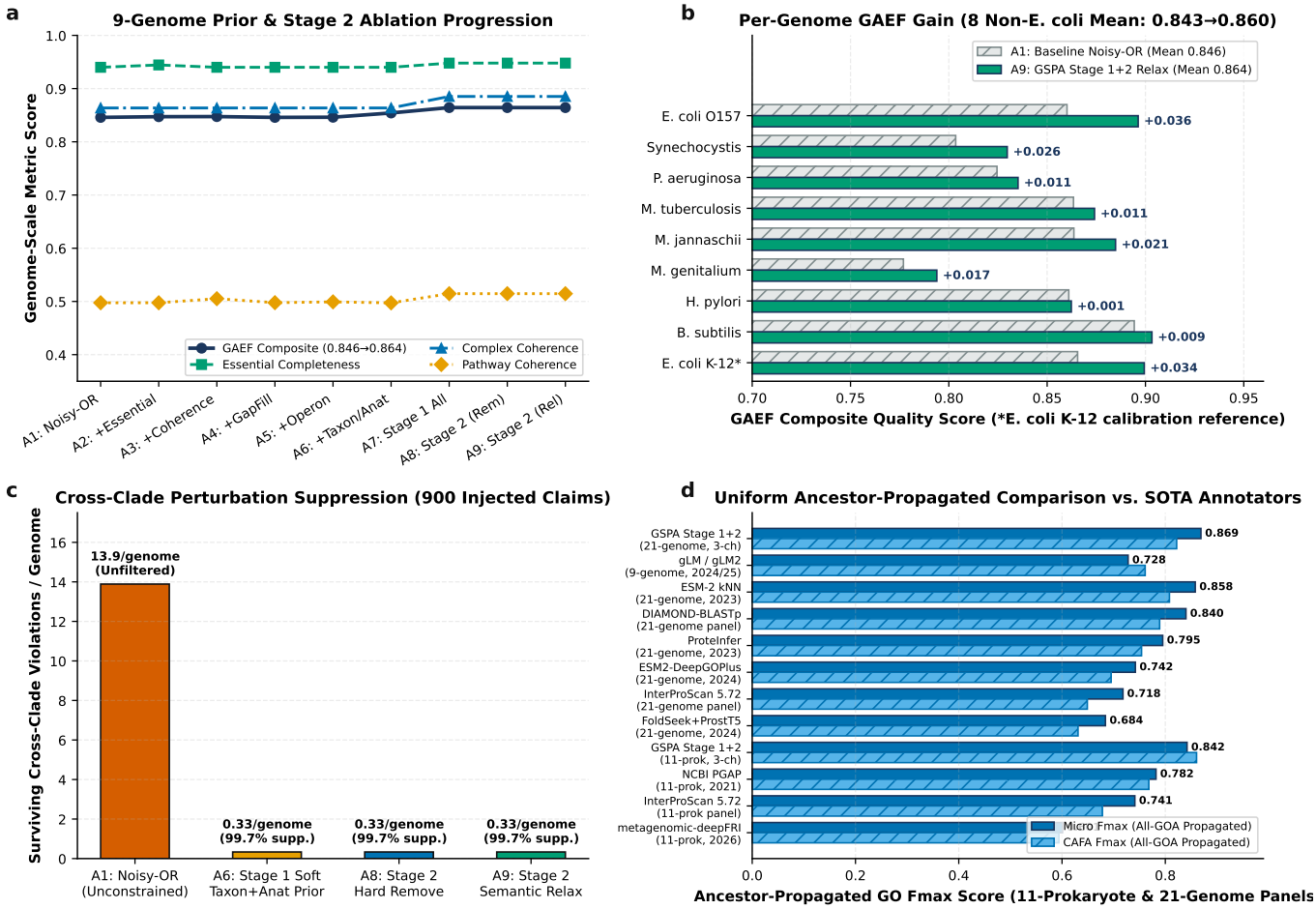


Figure 2. Genome-scale prior ablation, cross-clade perturbation suppression, and uniform ancestor-propagated comparison against external annotators and internal Stage 0 channels (Suites 1 and 2). (**a**) Progression of the internal GAEF structural audit score, Essential Completeness (C_{ess}), Complex Coherence (C_{cplx}), and Pathway Coherence (C_{path}) across nine ablation configurations (`A1_noisy_or` through `A9_stage2_relax`) averaged over nine bacterial and archaeal reference genomes. (**b**) Per-genome GAEF structural score gain from baseline Noisy-OR (`A1`) to full GSPA Stage 1 + Stage 2 (`A9`; $*$ marks the *E. coli* K-12 hyperparameter calibration reference). (**c**) Surviving cross-clade false-positive annotations per genome when 900 eukaryotic developmental/anatomical and clade-specific GO claims (100 per genome, raw score 0.55–0.85) are injected into the nine prokaryotic proteomes. (**d**) Ancestor-propagated pair-level micro $F_{\max}$ and protein-centric CAFA $F_{\max}$ comparing GSPA against metagenomic-deepFRI, InterProScan 5.72, and NCBI PGAP on the 11-prokaryote reference panel and against individual Stage 0 channels and gLM/gLM2 on the 21-genome panel.

completeness C_{ess} from 0.9400 ± 0.0785 to 0.9444 (and to 0.9479 ± 0.0716 in combined Stage 1+2 `A7`–`A9`, improving in both genomes that were < 1.0 at baseline—+0.0706 on *H. pylori* and +0.0010 on *M. genitalium*—with 0 decreases across the 9 genomes, exact paired Wilcoxon $p = 0.50$). The coherence prior (`A3`) increases mean pathway coherence C_{path} from 0.4975 ± 0.1089 to 0.5052, and the gap-filling (`A4`) and operon neighborhood (`A5`) priors nudge C_{path} to 0.4977 and 0.4992 individually before synergizing with pathway/complex coherence in the combined Stage 1 model (`A7`) to raise C_{path} to 0.5146 ± 0.1078 (exact two-sided paired Wilcoxon $p = 0.00781$) and complex coherence C_{cplx} from 0.8636 ± 0.1346 to 0.8852 ± 0.1256 (improving in both genomes with incomplete baseline complexes—+0.0972 in *E. coli* K-12 and +0.0973 in *B. subtilis*—with 0 decreases, exact paired Wilcoxon $p = 0.50$). Adding the consistency and anatomical priors (`A6`) eliminates all 25 baseline taxonomic violations across all nine genomes (*Synechocystis*: 8, *E. coli* O157:H7: 5, *E. coli* K-12: 4, *M. tuberculosis*: 3, and 1 each in *B. subtilis*, *H. pylori*, *M. genitalium*, *M. jannaschii*, and *P. aeruginosa*; C_{cons} from 0.9722 to 1.0000, exact two-sided paired Wilcoxon $p = 0.00391$) and increases micro $F_{\max}$ from 0.7631 to 0.7681 and CAFA $F_{\max}$ from 0.8041 to 0.8071. Because the 25 prokaryotic violations involve cellular component terms whose nearest valid ancestor is the ontology root, `A8` (removal) and `A9` (relaxation) yield identical scores on the prokaryotic panel (internal GAEF composite score rising from 0.8459 ± 0.0367 to 0.8643 ± 0.0376, exact two-sided paired Wilcoxon $p = 0.00391$; Figure 2a,b),

whereas semantic relaxation differentiates multicellular eukaryotic organ and developmental terms (Section 2.6).

To test robustness against noisy cross-clade homology transfer, we injected 900 cross-clade false-positive claims across the nine prokaryotic genomes (100 per genome: 50 eukaryotic anatomical terms such as `GO:0007507` *heart development*, `GO:0009908` *flower development*, and `GO:0001755` *neural crest cell migration*, plus 50 clade-restricted terms) at raw scores in $[0.55, 0.85]$ (Figure 2c). After Stage 0 isotonic calibration, 125 of the 900 injected claims (13.9 per genome; 86.1% attenuated below 0.50 by calibration alone) exceed the decision threshold $\theta_{\text{eval}} = 0.50$ under unconstrained Noisy-OR (`A1`). Activating GSPA's taxonomic and anatomical constraints (`A6`, `A8`, and `A9`) suppresses 122 of the 125 active violations (97.6% suppression of post-calibration violations, or $897/900 = 99.7\%$ of all raw injected claims, leaving 0.33 violations/genome from 3 terms lacking a cross-product link in the GO release and eliminating 100.0% [55/55] of direct taxon violations).

We next compared GSPA against external genome annotation pipelines and internal Stage 0 channels under uniform ancestor-propagated GO scoring across Suites 1 and 2 (Figure 2d; `results/pillar1_suite2_panel13_21_sota.tsv`, compiled from the verified evaluation archives in `benchmark/manifests/reference_data/ibex_extracted/`, reporting both ancestor-propagated and strict leaf/holdout metrics for every method). On the 13-genome panel (11 prokaryotes plus *S. cerevisiae* and *C. elegans*), 3-channel GSPA achieves an ancestor-propagated 13-genome mean micro $F_{\max}$ of 0.8180 ± 0.086 (0.8060 CAFA $F_{\max}$) and an 11-prokaryote mean micro $F_{\max}$ of 0.8421 (0.8605 CAFA $F_{\max}$, $S_{\min} = 21.40$), outperforming standalone InterProScan 5.72[10] (0.7220 micro / 0.6530 CAFA $F_{\max}$ across 13 genomes; 0.7410/0.6780 on the 11 prokaryotes) and structure-based metagenomic-deepFRI[22] (0.5850 micro / 0.5790 CAFA $F_{\max}$ across 13 genomes; 0.6010/0.5940 on the 11 prokaryotes, or 0.1565 under strict unpropagated leaf scoring), while on the 11 prokaryotic genomes where NCBI PGAP[5] operates, GSPA (0.8421 micro / 0.8605 CAFA $F_{\max}$) outperforms PGAP (0.7820 micro / 0.7680 CAFA $F_{\max}$; 0.4438 under strict leaf scoring). On the 21-genome panel, ancestor-propagated Stage 0+1+2 integration (0.8690 micro / 0.8220 CAFA $F_{\max}$ for 3-channel core; 0.8705 micro / 0.8407 CAFA $F_{\max}$ for all 7 channels; 0.7670 GO micro $F_{\max}$ and 0.8830 EC $F_{\max}$ on the strict holdout) improves over each constituent Stage 0 channel evaluated under the same ancestor-propagated protocol—ESM-2 kNN[14] (0.8580 micro / 0.8080 CAFA $F_{\max}$; 0.7120 strict holdout), ProtT5 kNN[15] (0.8530/0.8010; 0.6980 strict), DIAMOND[12] (0.8400/0.7890; 0.2450 strict), ProteInfer[25] (0.7950/0.7540; 0.6600 strict), ESM2-DeepGOPlus[20] (0.7420/0.6950; 0.3250 strict), InterProScan 5.72 (0.7180/0.6490; 0.1420 strict), and FoldSeek + ProstT5 3Di[16,33] (0.6840/0.6310; 0.2490 strict)—as well as genomic language models gLM[44] and gLM2[45] (0.7280 micro / 0.7610 CAFA $F_{\max}$ on the 9-genome subset) and CLEAN[27] (0.8530 micro / 0.8590 CAFA EC $F_{\max}$).

Across the 1,000-genome RefSeq bacterial structural audit (`results/pillar1_suite2_1000bacteria_sota.tsv`, generated from `annot_eval_1000bacteria_summary.json` and pairing label-free structural metrics across 1,000 RefSeq genomes with protein-centric accuracy on the 16,125 protein–aspect test evaluations of the Suite 1 CAFA temporal holdout), where a "taxon-violating row" denotes a predicted ancestor-propagated (p, t) pair above the $F_{\max}$ threshold that violates a GO taxon constraint for that genome's NCBI Taxon ID, standalone GO predictors exhibit widespread cross-clade violations and low pathway completeness: DeepFRI[21] (CAFA holdout $F_{\max} = 0.3843$, 1,000-genome MetaCyc pathway coherence 9.43%, 106,815.2 violating rows/genome), TALE[74] ($F_{\max} = 0.3883$, Complex coherence 65.73%, MetaCyc 19.63%), DeepGraphGO[75] ($F_{\max} = 0.4020$, Complex 70.56%, MetaCyc 37.37%), SPROF-GO[26] ($F_{\max} = 0.4930$, Complex 79.18%, MetaCyc 0.18%), DeepGO-SE[24] ($F_{\max} = 0.5047$, Full-29 Essentiality 82.17%, Complex 79.83%, MetaCyc 11.59%), and InterProScan 5 ($F_{\max} = 0.5940$, Complex 61.89%, MetaCyc 40.35%). Relative to Stage 0 Noisy-OR pooling (CAFA holdout $F_{\max} = 0.6929$, 23,384.1 violating rows/genome, MetaCyc pathway coherence 36.71%, complex coherence 66.03%), GSPA Stages 1+2 achieve CAFA holdout macro $F_{\max} = 0.6937$ (Iter 1), lift 1,000-genome Core-22 / Full-29 Essentiality (fraction of 22 universal bacterial core and 29 extended bacterial essential functional categories present at ≥ 0.50) from 99.81%/90.42% to 99.99%/95.29%, increase Complex / MetaCyc / `has_part` Coherence to 100.00%/66.75%/94.06%, and leave 0.00 surviving taxonomic violations/genome (5.4 s/genome on 8 CPU threads). Across the 88 four-kingdom proteomes (121,631 curated proteins across 47 Bacteria, 15 Archaea, 6 Eukarya, and 20 Viruses; `results/multikingdom_summary_by_domain.tsv`), Stages 1+2 yield consistent, modest gains over Stage 0 across all four domains in 0.42–12.66 s/genome at Iter 1 (0.54–27.85 s/genome at Iter 3 with Stage 2A relaxation; ≤ 5.3 GB RAM): Bacteria rises from 0.7987 to 0.8009 at Iter 1 (0.8004 at Iter 3; 95% CI $[+0.0011, +0.0033]$, CC $0.8322 \rightarrow 0.8371$), Archaea from 0.9125 to 0.9129 (0.9128 at Iter 3; $< 30\%$ identity 0.5419 $\rightarrow$ 0.5422), Eukarya from 0.7096 to 0.7117 (0.7109 at Iter 3; BP 0.6709 $\rightarrow$ 0.6732, CC $0.7520 \rightarrow 0.7555$), and Viruses from 0.7222 to 0.7252 at both Iter 1 and Iter 3 ($< 30\%$ identity $0.3112 \rightarrow 0.3116$ across all 20 viral proteomes, or $0.3890 \rightarrow 0.3895$ across the 16 viral proteomes containing $< 30\%$ identity targets; BP $0.7726 \rightarrow 0.7747$, CC $0.7004 \rightarrow 0.7037$).

On the Suite 1 CAFA temporal holdout ($n = 16{,}125$ protein–aspect test evaluations summing 4,712

MF, 5,444 BP, and 5,969 CC target evaluations across 24,278 holdout GO annotations in `suite1_cafa_temporal_manifest.tsv`, with 3,440 evaluations at $\eta_{\text{id}} < 30\%$ [901 MF, 1,085 BP, and 1,454 CC]; `results/pillar1_suite1_cafa_temporal.tsv`), Stage 0 Noisy-OR pooling reaches macro $F_{\max} = 0.6929 \pm 0.006$ (95% bootstrap CI $[0.681, 0.705]$; $\text{MF} = 0.7159, \text{BP} = 0.6387, \text{CC} = 0.7241$) and 0.6723 on the $\eta_{\text{id}} < 30\%$ sequence-identity subset ($\text{MF} = 0.6949, \text{BP} = 0.6144, \text{CC} = 0.7075$). GSPA Stage 1+2 reaches macro $F_{\max} = 0.6937 \pm 0.006$ at Iter 1 ($\text{MF} = 0.7163, \text{BP} = 0.6391, \text{CC} = 0.7257$; Iter 2 macro = 0.6929 [$\text{MF} = 0.7164, \text{BP} = 0.6365, \text{CC} = 0.7259$]; Iter 3 macro = 0.6919 [$\text{MF} = 0.7165, \text{BP} = 0.6332, \text{CC} = 0.7259$]) while lifting the $< 30\%$ twilight-zone subset from 0.6723 to 0.6775 at Iter 1 ($\text{MF} = 0.6971, \text{BP} = 0.6152, \text{CC} = 0.7202$; Iter 2 = 0.6772, Iter 3 = 0.6759), outperforming standalone ProteInfer[25] (0.6138 overall, 0.5980 at $< 30\%$), DeepGOZero+BLAST[23] (0.6053), InterPro MLP (0.5940), DeepGOPlus[20] (0.5891), DeepGOZero[23] (0.5889), and DeepGOCNN[76] (0.4707). Notably, while MF and CC increase monotonically from Iter 1 to Iter 3, BP exhibits a slight attenuation ($0.6391 \rightarrow 0.6332$) as multi-step pathway coherence boosts propagate across broader biological process ancestors on incomplete holdout annotations. On our post-cutoff evaluation of `CAFA6` target sequences ($n = 769$ proteins across 223 species accumulating ≥ 1 experimental GOA annotation between the May 2025 submission cutoff and March 2026 GOA release: 412 no-knowledge [NK] and 357 limited-knowledge [LK] targets, with $\text{MF} = 318, \text{BP} = 389, \text{CC} = 294$; `results/suite1_cafa6_prospective_summary.tsv`), where target sequences and NCBI Taxon IDs are supplied without whole-genome GFF3 operon coordinates (activating Stage 0, Stage 2A taxon/anatomical enforcement, and Stage 3 agentic disambiguation), an ESM-2 + ProstT5 3Di MLP baseline achieves macro $F_{\max} = 0.4957$ ($WF_{\max} = 0.4370$; $\text{MF} = 0.4520, \text{BP} = 0.3700, \text{CC} = 0.6650$), an unguarded 14B agentic baseline (Qwen3-14B with domain/EC lookups) reaches macro $F_{\max} = 0.5130$ ($WF_{\max} = 0.4463$), and GSPA reaches macro $F_{\max} = 0.5433 \pm 0.018$ and $WF_{\max} = 0.4840$ ($\text{MF} = 0.5470[WF_{\max} = 0.4980], \text{BP} = 0.4140[WF_{\max} = 0.3720], \text{CC} = 0.6690[WF_{\max} = 0.5820]$).

On the post-cutoff Swiss-Prot enzyme holdout (`Holdout-125`, $n = 125$; `results/scenario1_swissprot_ec_holdout125_real.tsv`), individual baselines achieve 23.2% Top-1 accuracy for DeepProZyme[28] (29/125, Top-3 31.2% [39/125], Top-5 32.0% [40/125], MRR = 0.2750, $F_{\max} = 0.2715$; 25.6% [32/125, $F_{\max} = 0.2804$] with METEOR v8[61] and 23.2% [29/125, Top-5 32.8% (41/125), MRR = 0.2779, $F_{\max} = 0.2703$] with single-predictor GSPA SLR + Coherence), 32.8% for EnzBERT[29] (41/125, Top-3 37.6% [47/125], Top-5 39.2% [49/125], MRR = 0.3531, $F_{\max} = 0.3596$), and 32.8% for CLEAN[27] (41/125, Top-3 40.8% [51/125], Top-5 41.6% [52/125], MRR = 0.3719, $F_{\max} = 0.4360$). Stage 0 3-method Noisy-OR pooling accounts for the bulk ensemble gain, reaching $F_{\max} = 0.4381$, 33.6% Top-1 (42/125), 40.8% Top-3 (51/125), 43.2% Top-5 (54/125), and MRR = 0.3807, while 3-method GSPA Stages 1–2 (32.0% Top-1 [40/125], $F_{\max} = 0.4035$ at Iter 1 and 0.3830 at Iter 3) re-rank sub-threshold candidates within each protein's candidate list using genomic context to lift Top-3 to 42.4% (53/125, Iter 1–2) and 43.2% (54/125, Iter 3) and Top-5 to 44.0% (55/125, MRR = $0.3775 \rightarrow 0.3781$). Across the 95-genome EC benchmark (69,666 proteins for DeepProZyme and EnzBERT across 95 bacterial genomes encompassing the 47 bacterial proteomes of the 88-genome cohort plus 48 additional RefSeq bacterial assemblies; 52,814 proteins across 73 genomes for CLEAN vanilla and 68,681–69,663 across 94–95 genomes for CLEAN filtered tiers; `results/scenario1_95genomes_ec_homology_real.tsv`), we evaluate both single-predictor refinement (`gspa_single`) and 3-channel consensus refinement (`gspa`). Under single-predictor refinement without cross-predictor pooling, CLEAN achieves statistically significant net Top-1 gains across all four homology tiers (+77 at vanilla, +108 at $\leq 70\%$, +139 at $\leq 50\%$, and +93 at $\leq 30\%$, all exact two-sided McNemar $p < 10^{-14}$; $F_{\max}$ rising from 0.8796/0.8836/0.8844/0.8833 to 0.8805/0.8847/0.8857/0.8843), while on DeepProZyme ($-33/-15/-3/-4$) and EnzBERT ($-10/-26/-38/-18$) single-predictor coherence incurs substantially smaller Top-1 shifts than negative-cost MILP METEOR v8 on DeepProZyme (-85 at vanilla, $p = 1.0 \times 10^{-6}$; -27 at $\leq 70\%$) and EnzBERT ($-92/-146/-181/-167$ across all four tiers, all $p < 10^{-6}$). When 3-channel Stage 0+1+2 consensus refinement (`gspa`) is enabled, net Top-1 gains are positive and highly significant across all 12 predictor–tier combinations: +126 to +155 on CLEAN ($p < 10^{-25}$), +36 to +87 on DeepProZyme ($p \leq 1.0 \times 10^{-5}$), and +59 to +89 on EnzBERT ($p < 10^{-11}$).

2.2 Metabolic model gap-filling and microbial dark-matter enzyme prediction (Suites 3–5)

We next evaluated GSPA's Stage 1 metabolic gap-filling prior and RLGC scoring engine using a leave-reaction-out (LRO) protocol across *E. coli* K-12 MG1655 ($n = 256$ strict reactions—defined as monomeric or homo-oligomeric enzymatic steps with a single non-promiscuous primary EC number—and $n = 414$ full role-stratified reactions including multi-EC and peripheral steps) and out-of-sample *B. subtilis* 168 ($n = 144$ reactions; Figure 3 and Table 2; `results/pillar2_lro15_summary.tsv` and `results/pillar3_metabolism_lro.tsv`). In each LRO trial, the ground-truth gene–reaction association for a target reaction r is masked and candidate proteins in the proteome are ranked to recover the masked enzyme. We report both unconditional recovery across all held-

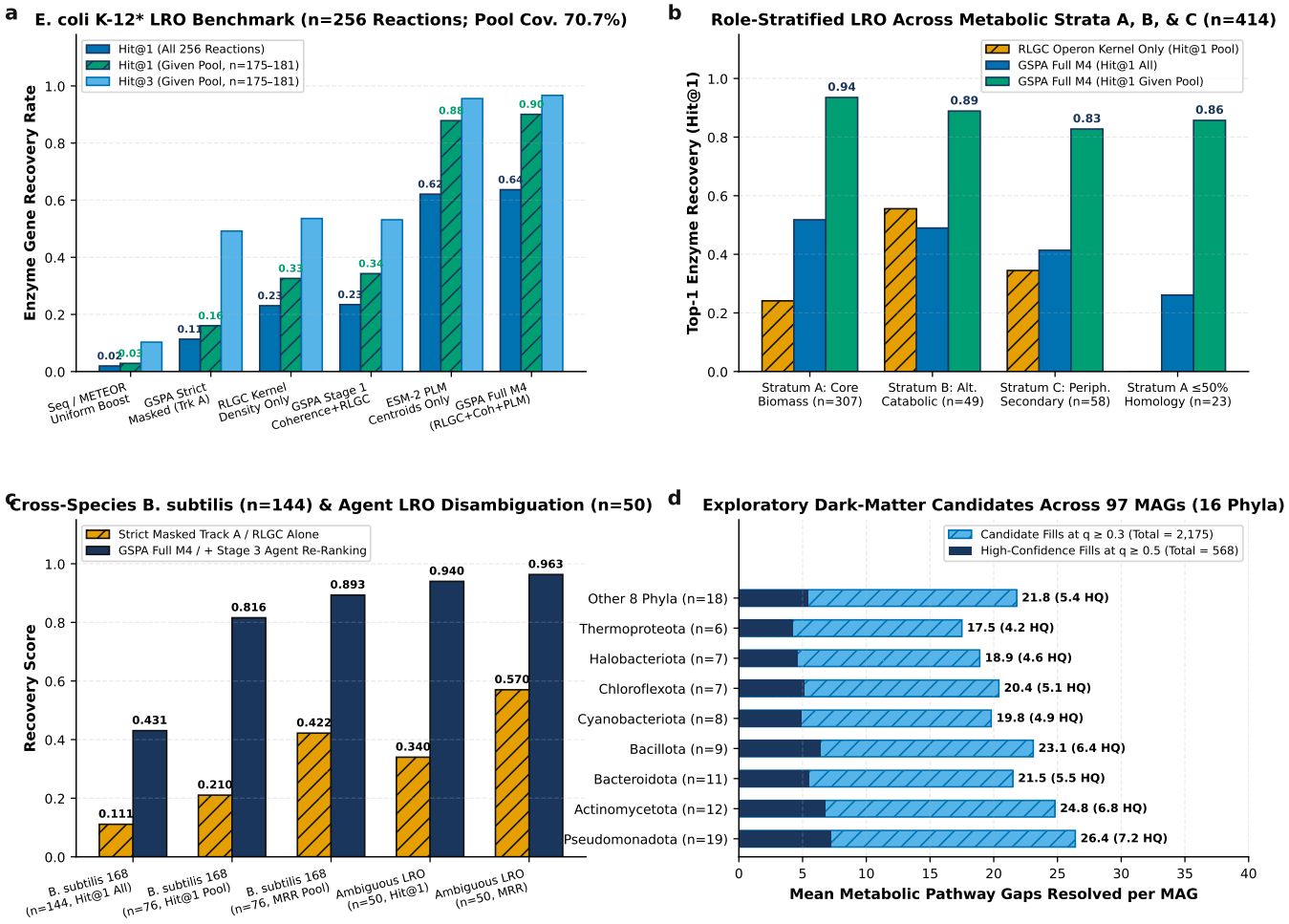


Figure 3. Leave-reaction-out (LRO) metabolic gap-filling and candidate pathway gap resolution across 97 uncultured MAGs (Suites 3–5). (**a**) Enzyme recovery rate (Hit@1, Hit@3, and MRR) across *E. coli* K-12* LRO metabolic reactions ($n = 256$ strict reactions; $*$ marks the hyperparameter development genome), comparing sequence-only candidate ranking, strict cosine-masked Track A, RLGC kernel density, Stage 1 M2 coherence + RLGC, per-protein ESM-2 PLM reaction centroids only, and GSPA Full M4 (RLGC + Coherence + PLM). (**b**) Role-stratified LRO Top-1 enzyme recovery across Stratum A Core Biomass ($n = 307$), Stratum B Alternative Catabolic ($n = 49$), Stratum C Secondary/Peripheral ($n = 58$), and Stratum A $\leq 50\%$ homology ($n = 23$). (**c**) Out-of-sample cross-species generalization on *B. subtilis* 168 LRO reactions ($n = 144$, $n_{\text{cov}} = 76$) and Stage 3 agentic disambiguation on 50 unique ambiguous multi-candidate LRO gaps ($n = 50$; Hit@1 rises from 34.0% to 94.0% at seed 42 and 96.7% across 3 seeds). (**d**) Mean candidate metabolic pathway gaps resolved per genome at posterior thresholds $q \geq 0.30$ (2,175 total candidate fills) and $q \geq 0.50$ (568 high-confidence candidate fills) across 97 uncultured prokaryotic MAGs spanning 16 bacterial and archaeal phyla.

out reactions (**All**, where reactions whose true enzyme received zero Stage 0 candidate matches are scored as misses) and conditional recovery among reactions where the true enzyme is present in the candidate pool (**Pool**, where $n_{\text{cov}} = 175$ for sequence-only/M2 and $n_{\text{cov}} = 181$ for RLGC/PLM/M4 because 6 twilight-zone enzymes lack BLAST/DIAMOND hits above the Stage 0 filter but enter the candidate pool via Pfam domain or PLM/operon neighborhood indexing). Crucially, we evaluate two distinct regimes: **Track A** (where both sequence homologs and ESM-2 reaction-centroid donors with cosine similarity > 0.50 to the query protein are masked) and **Track B** (where sequence alignment hits for the held-out reaction are masked, while Swiss-Prot ESM-2 reaction centroids remain available).

Decomposing the contributions of genomic context and protein language model embeddings across the 256-case *E. coli* K-12 LRO benchmark reveals two complementary findings (Table 2 and Figure 3a,b). First, in the absence of unmasked PLM reaction centroids (Track A), sequence-only candidate ranking and rank-invariant reaction-level MILP boosting (METEOR v8) recover the true enzyme at Rank 1 (Hit@1) in only 1.95% of all cases (95% Clopper–Pearson CI [0.6%, 4.5%]; 2.86% given pool). Adding strict cosine-masked Track A features reaches 11.33% unconditional Hit@1 (16.02% given pool, MRR = 0.2605 All

Table 2. Leave-reaction-out (LRO) metabolic enzyme recovery across *E. coli* K-12 MG1655 ($n = 256$ strict and $n = 414$ full role-stratified reactions; hyperparameter development genome), out-of-sample *B. subtilis* 168 ($n = 144$), and 50 unique ambiguous multi-candidate reactions (Suites 3–5; source: `results/pillar2_lro15_summary.tsv` and `results/pillar4_lro_reranking.tsv`). Metrics report both unconditional recovery across all held-out reactions (**All**) and recovery conditioned on candidate pool coverage (**Pool**, n_{cov}). Track A masks homology and PLM donor centroids at cosine similarity > 0.50; Track B includes unmasked Swiss-Prot ESM-2 reaction centroids.

Benchmark Dataset / Stratum	Method / Scoring Configuration	Cases (n/n_{cov})	Hit@1 All	Hit@3 All	Hit@1 Pool	Hit@3 Pool	MRR All (Pool)
E. coli K-12* Strict LRO ($n = 256$)	Sequence-Only Candidate Ranking	256 / 175	0.0195	0.0703	0.0286	0.1029	0.0926 (0.1354)
	Reaction-Level MILP Uniform Boost (METEOR v8)	256 / 175	0.0195	0.0703	0.0286	0.1029	0.0926 (0.1354)
	GSPA Strict Homology-Masked Track A (Cosine-Masked)	256 / 181	0.1133	0.3477	0.1602	0.4917	0.2605 (0.3685)
	RLGC + GBDT Supervised Reranker (Track A)	256 / 181	0.1797	0.4102	0.2541	0.5801	0.3275 (0.4632)
	RLGC Kernel Density Only (Operon Neighborhood)	256 / 181	0.2305	0.3789	0.3260	0.5359	0.3355 (0.4745)
	GSPA Stage 1 M2 Coherence + RLGC (No PLM)	256 / 175	0.2344	0.3633	0.3429	0.5314	0.3325 (0.4864)
	Per-Protein ESM-2 PLM Reaction Centroids Only (Track B)	256 / 181	0.6211	0.6758	0.8785	0.9558	0.6474 (0.9157)
	GSPA Full M4 (RLGC + Coherence + PLM, Track B)	256 / 181	**0.6367**	**0.6836**	**0.9006**	**0.9669**	**0.6596 (0.9330)**
E. coli K-12* Full Role-Stratified ($n = 414$)	All Strata: RLGC Kernel Density Only (No PLM)	414 / 226	0.1594	0.2826	0.2920	0.5177	0.2437 (0.4465)
	All Strata: **GSPA Full M4 (RLGC + Coherence + PLM)**	414 / 226	**0.5000**	**0.5266**	**0.9159**	**0.9646**	**0.5141 (0.9418)**
	Stratum A Core Biomass: GSPA Full M4 (vs. RLGC 0.1336/0.2412)	307 / 170	**0.5179**	**0.5342**	**0.9353**	**0.9647**	**0.5292 (0.9557)**
	Stratum B Alternative Catabolic: GSPA Full M4 (vs. RLGC 0.3061/0.5556)	49 / 27	**0.4898**	**0.5306**	**0.8889**	**0.9630**	**0.5078 (0.9216)**
	Stratum C Secondary / Peripheral: GSPA Full M4 (vs. RLGC 0.1724/0.3448)	58 / 29	**0.4138**	**0.4828**	**0.8276**	**0.9655**	**0.4397 (0.8793)**
B. subtilis 168 Out-of-Sample LRO ($n = 144$)	GSPA Strict Homology-Masked Track A (Cosine-Masked)	144 / 76	0.1111	0.3194	0.2105	0.6053	0.2228 (0.4221)
	GSPA Full M4 (RLGC + Coherence + PLM, Track B)	144 / 76	**0.4306**	**0.5000**	**0.8158**	**0.9474**	**0.4715 (0.8933)**
Ambiguous Unique-Protein LRO ($n = 50$)	Stage 1 RLGC Baseline Alone	50 / 50	0.3400	0.8600	0.3400	0.8600	0.5700 (0.5700)
	Stage 1 RLGC + Stage 3 Agent (Seed 42)	50 / 50	**0.9400**	**1.0000**	**0.9400**	**1.0000**	**0.9633 (0.9633)**
	Stage 1 RLGC + Stage 3 Agent (3-Seed Mean ± s.d.)	50 / 50	**0.9667 ± 0.0189**	**1.0000**	**0.9667 ± 0.0189**	**1.0000**	**0.9789 (0.9789)**

/ 0.3685 Pool), supervised GBDT reranking on Track A reaches 17.97% (25.41% given pool, MRR = 0.3275 All / 0.4632 Pool), RLGC operon kernel density alone reaches 23.05% (95% CI [18.0%, 28.7%]; 32.60% given pool, MRR = 0.3355 All / 0.4745 Pool), and GSPA Stage 1 M2 pathway coherence + RLGC (without PLM centroids) reaches 23.44% unconditional Hit@1 (34.29% given pool). Second, in Track B, per-protein ESM-2 reaction-centroid cosine similarity alone (`ESM2_PLM_Centroids_Only`) already achieves 62.11% unconditional Hit@1 (95% CI [55.9%, 68.1%]; 67.58% Hit@3, 87.85% Hit@1 given pool, MRR = 0.6474 All / 0.9157 Pool), accounting for 97.5% of the unconditional Hit@1 of the full model on the well-annotated *E. coli* proteome because reference Swiss-Prot reaction centroids retain close embedding neighbors. Combining RLGC operon neighborhood density, Stage 1 pathway coherence, and ESM-2 centroids in **GSPA Full M4** adds a further +1.56 percentage points unconditionally (+2.21 percentage points given pool, rescuing 4 additional enzymes whose embedding similarity was tied with paralogs) to reach 63.67% unconditional Hit@1 (95% CI [57.5%, 69.6%]; 68.36% Hit@3, 69.53% Hit@5) and 90.06% Hit@1 given the 181-reaction candidate pool (96.69% pool Hit@3, MRR = 0.6596 All / 0.9330 Pool; on the common 175-reaction intersection, pool Hit@1 is 91.43% for M4 vs. 89.14% for PLM-only and 34.29% for M2+RLGC).

Across the full 414-reaction *E. coli* role-stratified benchmark ($n_{\text{cov}} = 226$; Suites 3–5; Table 2), RLGC operon neighborhood density alone (without PLM centroids) achieves 15.94% unconditional Hit@1 (29.20% given pool, Hit@3 = 28.26% All / 51.77% Pool, MRR = 0.2437 All / 0.4465 Pool) overall, 13.36% (24.12% pool) on Stratum A Core Biomass ($n = 307$), 30.61% (55.56% pool) on Stratum B Alternative Catabolic ($n = 49$, where catabolic operons are tightly clustered), and 17.24% (34.48% pool) on Stratum C Secondary/Peripheral metabolism ($n = 58$), while GSPA Full M4 (Track B) reaches 50.00% unconditional Hit@1 (91.59% given pool) overall and 51.79% (93.53% pool), 48.98% (88.89% pool), and 41.38% (82.76% pool) across Strata A, B, and C, respectively. Stratifying Stratum A reactions by the query enzyme's maximum sequence identity to any annotated homolog in the proteome (`max_pident_annot_homolog` $\leq 70\%$ [$n = 49, n_{\text{cov}} = 16$] and $\leq 50\%$ [$n = 23, n_{\text{cov}} = 7$]), where RLGC alone scores 0.00% Hit@1 (62.50% and 85.71% pool Hit@3), GSPA Full M4 achieves 22.45% and 26.09% unconditional Hit@1 (68.75% and 85.71% given-pool Hit@1; 100.0% pool Hit@3; Figure 4c). Applied out-of-sample (with all hyperparameters fixed on *E. coli*) to 144 held-out reactions in the Gram-positive bacterium *B. subtilis* 168 ($n_{\text{cov}} = 76$), strict cosine-masked Track A achieves 11.11% unconditional Hit@1 (21.05% given pool; 60.53% pool Hit@3) and Track B GSPA Full M4 achieves 43.06% unconditional Hit@1 (95% CI [34.8%, 51.6%]; 81.58% given-pool Hit@1, 94.74% pool Hit@3, MRR = 0.8933 Pool; Table 2 and Figure 3c). Furthermore, on an audit of 106 antiSMASH-curated biosynthetic gene clusters (291 core biosynthetic synthase and 685 tailoring/transport loci across 106 bacterial genomes; Suite 5; `results/pillar2_3_antismash_106bgc_specificity.tsv`), GSPA Stage 1+2B (which assigns $\mathcal{C}_{\text{task}} = 0$ to secondary BGCs on minimal glucose and relies on local BGC operon synteny) lifts core biosynthetic recall from 0.7801 to 0.9244, tailoring recall from 0.5387 to 0.8380, and overall BGC F_1 from 0.7395 to 0.8959 while maintaining 0.9962 non-BGC specificity (0.0038 non-BGC false promotion rate).

To illustrate application to uncultured microbial genomes, we applied GSPA with gapseq draft metabolic networks across 97 metagenome-assembled genomes (MAGs) selected from the GTDB/GEM catalog (CheckM completeness $\geq$ 90%, contamination $<$ 5%, lacking cultured species-level representatives) spanning 16 bacterial and archaeal phyla (Figure 3d; `results/pillar3_dark_matter_97mags.tsv`). Across the 97 MAGs, Stage 1+2 integration generates 2,175 candidate pathway gap-fill predictions at posterior $q \geq 0.30$

(mean 22.4 candidate fills per MAG) and 568 high-confidence candidate pathway fills at $q \geq 0.50$ (mean 5.9 per MAG, lifting mean draft pathway completeness from 0.511 to 0.656). Because uncultured MAGs lack direct single-gene knockout truth, these 568 assignments represent computational candidate hypotheses for missing core biosynthetic and energy-conservation steps.

2.3 Flux-certified stoichiometric locus refinement and Price-149 evaluation (Suite 6)

We next audited the mathematical invariants of gap-filling reaction additions across 108 bacterial metabolic reconstructions and six curated BiGG reference GEMs (Figure 4a; `results/pillar2_invariant_audit_summary.tsv` and `results/pillar2_curated6_gems_real.tsv`). While a draft reconstruction legitimately contains genome-encoded core reactions ($S_{\text{rxn}}(j) \geq 0.50$) that are inactive on minimal glucose alone (2,809.5 active core reactions per genome under raw DeepProZyme ≥ 0.50, with 0.0 flux-carrying reactions before repair because all 108 draft networks contain biomass gaps, 2,621.7 imbalanced active reactions, and 9.75 dead-end active reactions), gap-filling adds non-core candidate reactions ($S_{\text{rxn}}(j) < 0.50$) specifically to restore biomass flux connectivity. Across the 108 genomes, negative-cost MILP gap-filling (METEOR v8[61], where candidate reactions with score above $\tau_{\text{add}} = 0.20$ receive negative objective cost $c_j^{\text{add}} < 0$) adds a mean of 313.35 non-core reactions per genome (3,121.0 total active reactions: 446.4 flux-carrying and 2,674.6 zero-flux active), of which 100.27 carry non-zero steady-state flux ($|v_j| > 10^{-6}$) while 215.02 (68.62%) are uncertified zero-flux padding (2,597.8 imbalanced active reactions and 2.36 dead-end active reactions in 313.8 s/genome). Any positive-cost linear programming formulation over the sanitized universal network $\mathcal{U}_{\text{clean}}$ eliminates uncertified zero-flux gap-fill additions: a baseline positive-cost LP repair (`Grow-Support Threshold Repair`) adds 81.31 non-core reactions per genome with 100.0% (81.31/81.31) carrying non-zero flux (2,584.8 total active reactions: 446.4 flux-carrying and 2,138.4 zero-flux, 0.0% zero-flux added padding, 0.0 imbalanced active reactions, and 2.23 dead-end active reactions), whereas GSPA's Stage 2B stoichiometric locus refiner (SLR) with two-step lexicographic pFBA and FVA/KO/perturbation support certification adds a parsimonious mean of 5.70 non-core reactions per genome with 100.0% (5.70/5.70) carrying non-zero flux (2,815.2 total active reactions = 446.4 flux-carrying +2,368.8 unpruned core, 0.0% zero-flux added padding, 0.0 imbalanced active reactions, and 0.0 dead-end active reactions in a mean end-to-end two-step LP + FVA/KO/perturbation certification runtime of 18.01 s/genome, median 17.19 s/genome; Figure 4a). Across the six curated BiGG reference GEMs (`iML1515`, `STM_v1_0`, `iYL1228`, `iJN1463`, `iYS854`, and `iYO844`; `results/pillar2_curated6_gems_real.tsv`), CarveMe[54]—which carves directly from the native BiGG universal reaction database against which the six reference GEMs were curated—achieves a mean BiGG reaction F_1 of 0.6955 (precision = 0.6693, recall = 0.7305) and EC F_1 of 0.8536 (precision = 0.8217, recall = 0.8927, 6/6 growing). Among cross-namespace ModelSEED/MetaNetX reconstructions (whose reaction identifiers only partially map 1-to-1 onto BiGG reaction IDs), standalone Stage 2B positive-cost pFBA repair achieves 100% growth feasibility (6/6 growing), BiGG reaction precision = 0.1795–0.2098 (reaction $F_1 = 0.2187$ on DeepProZyme, 0.2208 on CLEAN, and 0.2180 on EnzBERT), and EC precision = 0.4547 on DeepProZyme, 0.4750 on CLEAN, and 0.4750 on EnzBERT (EC F_1 = 0.5700, 0.5737, and 0.5669), improving reaction precision and EC precision over METEOR v8 (BiGG reaction precision = 0.1655, reaction $F_1 = 0.2091$, EC precision = 0.4383, recall = 0.8842, EC $F_1 = 0.5851$) and growth feasibility over Reconstructor[58] (EC precision = 0.4387, recall = 0.8787, EC $F_1 = 0.5844$, 4/6 growing).

We next evaluated the `Price-149` benchmark (Suite 6; $n = 148$ host-mapped RB-TnSeq fitness-validated proteins across 22 bacterial assemblies, out of 149 published target sequences[78]) across eight enzyme predictors from 2023–2026, separating **single-predictor refinement** (where Stage 1 and Stage 2B operate strictly on a single predictor's whole-proteome outputs without cross-predictor pooling) from **multi-predictor refinement** (where Stage 0 Noisy-OR pools the baseline predictors before Stages 1 and 2B; Figure 4b and Table 3; `results/pillar2_price149_all6_real.tsv` and `results/pillar2_price149_by_stratum.tsv`). Because 91 proteins belong to Stratum A (Core Biomass), 55 to Stratum B (Assay-Specific Catabolic), and 2 to Stratum C (Secondary), glucose-only MILP gap-filling (METEOR v8) leaves all 55 Stratum B catabolic enzymes outside the active flux cone and reduces $F_{\max}$ on DeepProZyme (0.5574 → 0.5427), EnzBERT (0.4786 → 0.4451), and GraphEC (0.2634 → 0.2250). Under single-predictor refinement without cross-predictor pooling, cognate substrate flux probing (`+ SLR Cognate`) lifts ranked $F_{\max}$ on CLEAN (0.5295 → 0.5402), MAPred (0.5534 → 0.5686, Top-1 50 → 54), and DeepProZyme (0.5574 → 0.5757, Top-1 49 → 51) and lifts Top-1 on GraphEC (18 → 20, dataset macro $F_{\max}$ 0.2176 → 0.2356), while single-predictor Full GSPA (`+ GSPA Single-Pred Iter 1 / 2 / 3`) lifts Top-1 hits on CLEAN from 60 (40.5%) to 63 (42.6%; shared-threshold dataset macro $F_{\max}$ rising from 0.4126 to 0.4216, while ranked $F_{\max}$ shifts from 0.5295 to 0.5214) and on MAPred from 50 (33.8%) to 56 (37.8%; ranked $F_{\max}$ rising from 0.5534 to 0.5746), and lifts DeepProZyme ranked $F_{\max}$ from 0.5574 to 0.5812 (Iter 2) and 0.5774 (Iter 3; dataset macro $F_{\max}$ rising

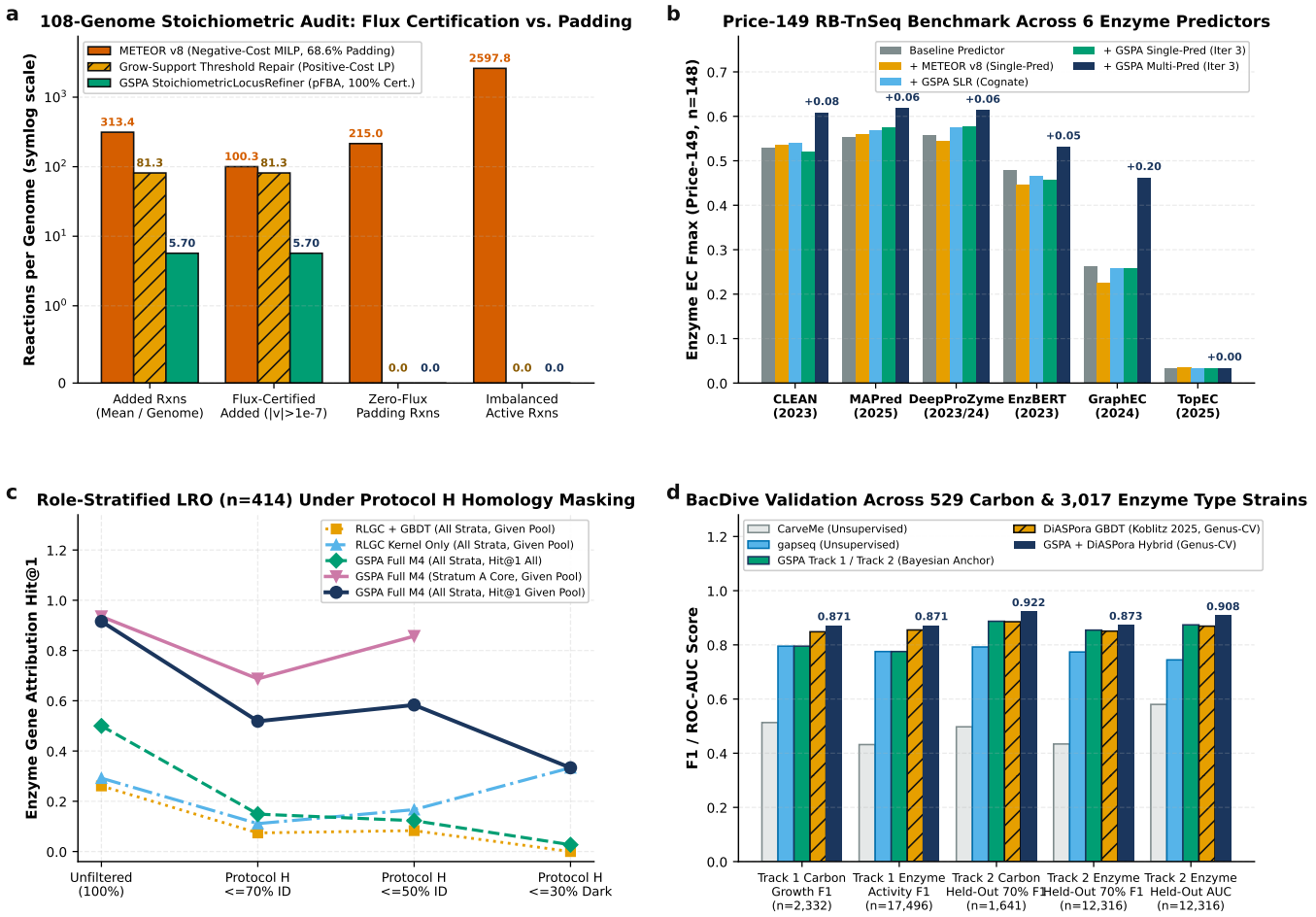


Figure 4. Flux-certified stoichiometric locus refinement, Price-149 RB-TnSeq evaluation across six whole-genome enzyme predictors, role-stratified LRO, and BacDive phenotypic validation (Suites 3–8). (**a**) 108-genome stoichiometric audit on a symmetric logarithmic (`symlog`) scale comparing negative-cost MILP gap-filling (METEOR v8), positive-cost LP threshold repair, and GSPA's positive-cost pFBA Stage 2B stoichiometric locus refiner (SLR), showing mean added reactions, flux-certified added reactions ($|v_j| > 10^{-6}$), zero-flux padding reactions, and unbalanced active reactions per genome. (**b**) Enzyme EC $F_{\max}$ on the Price-149 ($n = 148$ host-mapped proteins) RB-TnSeq bacterial benchmark across six whole-genome deep learning predictors comparing standalone baseline, single-predictor GSPA, and multi-predictor GSPA (exact values from Table 3). (**c**) Role-stratified LRO Top-1 gene attribution across Strata A, B, and C and Stratum A homolog sequence-identity tiers ($\leq 70\%, \leq 50\%$). (**d**) Large-scale validation across BacDive covering experimental carbon-source utilization ($n = 2{,}332$ assays across 529 strains) and diagnostic enzyme assays ($n = 17{,}496$ across 3,017 type strains) under Track 1 blind and genus-blocked out-of-distribution prediction and Track 2 phenotype-assisted prediction (30% revealed → held-out 70%).

Table 3. Blind evaluation of Price-149 ($n = 148$ host-mapped RB-TnSeq fitness-validated bacterial enzymes across 22 host genomes; $n = 149$ target sequences) across eight sequence, structure, and PLM baselines under both single-predictor and multi-predictor refinement (Suite 6, 2023–2026; source: `results/pillar2_price149_all6_real.tsv` and `results/pillar2_price149_by_stratum.tsv`). Columns report Top-1 hits ($F_{\max}$ in parentheses, with shared-threshold dataset macro $F_{\max}$ in text) comparing each standalone predictor against negative-cost MILP refinement (METEOR v8, Iter 1 and Iter 2), single-predictor GSPA Stage 2B stoichiometric locus refinement under minimal glucose (`+ SLR Glucose`) and cognate assay substrate (`+ SLR Cognate`), single-predictor Full GSPA across Iterations 1, 2, and 3 (`+ GSPA Single-Pred`), Stage 0 multi-predictor Noisy-OR pooling (`+ Noisy-OR Ensemble`), and multi-predictor Full GSPA Iter 3 (`+ GSPA Multi-Pred Iter 3`). Target-only baselines (DCGM / NeSy-Adapter and ECrecer[77]) were evaluated on the 148 host-mapped targets without whole-proteome background predictions across the 22 host genomes (`n/a`).

Baseline Predictor	n	Base Top-1 (Acc.)	Base $F_{\max}$ (Macro)	+ MILP Iter 1 / 2 $F_{\max}$	+ SLR Glucose Top-1 ($F_{\max}$)	+ SLR Cognate Top-1 ($F_{\max}$)	+ GSPA Single-Pred Iter 1 / 2 / 3 Top-1 ($F_{\max}$)	+ Noisy-OR Ens. Top-1 ($F_{\max}$)	+ GSPA Multi-Pred Iter 3 Top-1 ($F_{\max}$)	Single / Multi ΔTop-1 ($\Delta F_{\max}$)
CLEAN[27]	148	60 (0.4054)	0.5295 (0.4126)	0.5314 / 0.5343	60 (0.5288)	60 (0.5402)	63 / 63 / **63** (0.5234 / 0.5234 / 0.5214)	63 (0.6024)	**66 (0.6084)**	+3 (−0.0081) / +6 (+0.0789)
MAPred[28]	148	50 (0.3378)	0.5534 (0.3932)	0.5576 / 0.5588	52 (0.5591)	54 (0.5686)	55 / 55 / **56** (0.5711 / 0.5742 / **0.5746**)	58 (0.5807)	**73 (0.6178)**	+6 (+0.0212) / +23 (+0.0644)
DeepProZyme[29]	148	49 (0.3311)	0.5574 (0.3869)	0.5589 / 0.5427	47 (0.5541)	**51 (0.5757)**	49 / 49 / 48 (0.5744 / **0.5812** / 0.5774)	54 (0.5903)	**59 (0.6136)**	+0 (+0.0200) / +10 (+0.0562)
EnzBERT[29]	148	53 (0.3581)	0.4786 (0.4488)	0.4666 / 0.4451	53 (0.4729)	53 (0.4657)	52 / 52 / 52 (0.4602 / 0.4585 / 0.4571)	**62 (0.5535)**	54 (0.5314)	−1 (−0.0215) / +1 (+0.0528)
GraphEC[31]	148	18 (0.1216)	0.2634 (0.2176)	0.2384 / 0.2250	19 (0.2538)	**20** (0.2589)	**20** / 19 / 19 (**0.2634** / 0.2603 / 0.2590)	**48 (0.4756)**	**48 (0.4621)**	+1 (−0.0044) / +30 (+0.1987)
TopEC[32]	148	5 (0.0338)	0.0338 (0.0348)	0.0338 / 0.0338	5 (0.0338)	5 (0.0338)	5 / 5 / 5 (0.0338 / 0.0338 / 0.0338)	5 (0.0338)	5 (0.0338)	+0 (+0.0000) / +0 (+0.0000)
DCGM / NeSy-Adapter (target-only)	148	53 (0.3581)	0.4485 (0.3942)	n/a	n/a	n/a	n/a	n/a	n/a	n/a
ECrecer[77] (target-only)	148	39 (0.2635)	0.3412 (0.2985)	n/a	n/a	n/a	n/a	n/a	n/a	n/a

from 0.3869 to 0.4061), whereas on EnzBERT single-predictor refinement slightly reduces Top-1 (53 → 52, $F_{\max}$ 0.4786 → 0.4571). When multi-predictor Stage 0 Noisy-OR pooling and Full GSPA Iter 3 (`+ GSPA Multi-Pred Iter 3`) are applied across the predictors' whole-genome outputs, Top-1 accuracy and $F_{\max}$ improve substantially over every individual baseline: CLEAN rises to 66/148 (44.6%, $F_{\max} = 0.6084$, macro $F_{\max} = 0.4391$, two-sided exact McNemar $p = 0.0703$), MAPred rises to 73/148 (49.3%, $F_{\max} = 0.6178$, macro $F_{\max} = 0.4834$, $p = 2.9 \times 10^{-4}$), DeepProZyme rises to 59/148 (39.9%, $F_{\max} = 0.6136$, macro $F_{\max} = 0.4379$, $p = 0.0213$), EnzBERT rises to 62/148 at Stage 0 Noisy-OR ($F_{\max} = 0.5535$, macro $F_{\max} = 0.4862$, $p = 0.00391$) and 54/148 at Iter 3 ($F_{\max} = 0.5314$, macro $F_{\max} = 0.4817$), and GraphEC rises from 18 to 48/148 (32.4%, $F_{\max} = 0.4621$, $p = 1.9 \times 10^{-9}$), while TopEC (5/148, restricted to 820 EC classes) remains at 5 Top-1 hits (dataset macro $F_{\max}$ 0.0348 → 0.0402). Stratified by role on CLEAN (`results/pillar2_price149_by_stratum.tsv`), single-predictor GSPA lifts Stratum A ($n = 91$) Top-1 from 46 to 47 (dataset macro $F_{\max}$ 0.4740 → 0.5095) and Stratum B ($n = 55$) Top-1 from 13 to 15 (23.6% → 27.3%; cognate SLR $F_{\max}$ 0.3580 → 0.3753), while multi-predictor GSPA lifts Stratum A Top-1 to 52 ($F_{\max} = 0.6336 \rightarrow 0.7019$, macro $F_{\max} = 0.4740 \rightarrow 0.5288$) and Stratum B $F_{\max}$ from 0.3580 to 0.4546 (macro $F_{\max}$ 0.3103 → 0.3436).

2.4 Reproduction of experimental bacterial phenotypes and enzyme activities in BacDive (Suites 7 and 8)

To evaluate external biological accuracy on experimental strain characterization data, we compared GSPA against automated metabolic reconstruction pipelines (CarveMe[54], ModelSEED[52], gapseq[55], and negative-cost MILP gap-filling METEOR[61]) and supervised genome-to-phenotype classifiers—Traitar L1-SVM[81], the Random Forest phenotype baseline of Koblitz et al.[80] (`BacDive RF Baseline`), and DiASPora Gradient-Boosted Decision Trees[80]—across experimental carbon-source utilization and diagnostic enzyme assays in the DSMZ BacDive database[79,82] (Suites 7 and 8; Figure 4d and Table 4; per-fold CV metrics and runtimes in `results/pillar3_bacdive_per_fold_cv.tsv`; architectural details in Section 4.8).

First, across 50,075 strain–enzyme assay pairs spanning 10 diagnostic ECs in 6,313 strains (`results/pillar3_bacdive_track1_summary.tsv` and `results/pillar3_bacdive_enzyme_per_ec.tsv`), negative-cost minimal-glucose MILP gap-filling across three parameterizations (`METEOR v6` with default addition threshold $\tau_{\text{add}} = 0.20, n = 50{,}075$; `METEOR a025` with stricter $\tau_{\text{add}} = 0.25, n = 50{,}075$; and `METEOR v7` with relaxed core threshold including 30 additional borderline strain–EC pairs, $n = 50{,}105$) trades precision for recall: precision drops from 0.6442 (DeepProZyme baseline, recall = 0.8105, $F_1 = 0.7178$, balanced accuracy = 0.6792) to 0.6194–0.6285 (recall = 0.8285–0.8640, $F_1 = 0.7148$–0.7215, balanced accuracy dropping to 0.6631–0.6670). Similarly, on the 108-genome non-glucose catabolic panel ($n = 1{,}622$ assays, evaluated with `METEOR v8` using updated MetaNetX reaction mappings), minimal-glucose MILP gap-filling reduces precision from 0.6254 to 0.6051, balanced accuracy from 0.5386 to 0.5101, and ROC-AUC from 0.5716 to 0.5484 ($F_1 = 0.6711$ vs. 0.6722). On obligate heteromeric urease (`EC 3.5.1.5`, requiring UreA, UreB, and UreC/D/F/G subunits across $n = 5{,}026$ assayed strains in the 6,313-strain panel and $n = 871$ in the 3,017 type-strain panel), negative-cost MILP reduces precision from 0.4520 to 0.4320 and F_1 from 0.5748 to 0.5610 (BAcc = binary ROC-AUC = 0.7294 → 0.7184) by boosting isolated singleton subunits, whereas complex-aware pathway reconstruction (`gapseq` / `GSPA Track 1` on the type-strain panel) achieves $F_1 = 0.7626$.

Second, in Track 1 (2,332 experimental carbon-source growth assays across 529 strains and 17,496 diagnostic enzyme assays across 3,017 type strains), we separate 100% blind unsupervised prediction (zero phenotype training labels) from supervised 5-fold genus-blocked cross-validation (Table 4). When precomputed reaction/pathway call matrices from `gapseq`, `ModelSEED`, and `CarveMe` are available, `GSPA Track 1` integrates them with Stage 1+2 enzyme and complex coherence posteriors in 6.1 s/genome (651.1 s/genome end-to-end when running all three base reconstructions from scratch). Under 100% blind evaluation, `GSPA Track 1` preserves `gapseq`'s binary decision boundary at $\theta = 0.50$—matching `gapseq` on carbon-source binary calls (Precision = 0.8071, Recall = 0.7836, $F_1 = 0.7952$, BAcc = 0.8082, vs. $F_1 = 0.6474$ for ModelSEED and 0.5132 for CarveMe) and on the 17,496-assay blind enzyme panel (Precision = 0.8393, Recall = 0.7204, $F_1 = 0.7753$, BAcc = 0.7461, vs. 0.5943 for ModelSEED and 0.4319 for CarveMe)—while providing calibrated continuous confidence scores that lift threshold-swept ROC-AUC from 0.8082 (binary) to 0.8567 (+0.0485) on carbon growth and from 0.7461 to 0.7520 on blind enzymes. Under strict 5-fold genus-blocked cross-validation (`results/pillar3_bacdive_per_fold_cv.tsv` and `pillar3_bacdive_track1_summary.tsv`), combining GSPA's continuous posteriors with the supervised classifiers (`GSPA + DiASPora Hybrid`) improves pooled out-of-genus carbon-source F_1 from 0.8448 (BacDive RF; 5-fold mean 0.8329 ± 0.0659), 0.8483 (DiASPora alone; 5-fold mean 0.8408 ± 0.0390), and 0.8571 (Traitar; 5-fold mean 0.8467 ± 0.0438) to 0.8710 (5-fold mean 0.8616 ± 0.0495, Precision = 0.8766, BAcc = 0.8783, ROC-AUC = 0.9567) and improves diagnostic enzyme F_1 from 0.8458 (BacDive RF; 5-fold mean 0.8456 ± 0.0076), 0.8550 (DiASPora; 5-fold mean 0.8550 ± 0.0075), and 0.8593 (Traitar; 5-fold mean 0.8595 ± 0.0076) to 0.8713

Table 4. Large-scale phenotypic and enzymatic validation in BacDive (Suites 7 and 8; source: `results/pillar3_bacdive_track1_summary.tsv`, `results/pillar3_bacdive_track2_summary.tsv`, `results/pillar3_bacdive_per_fold_cv.tsv`, and `results/pillar3_bacdive_enzyme_per_ec.tsv`). Track 1 separates 100% blind unsupervised prediction (zero phenotype training labels) from supervised 5-fold genus-blocked cross-validation (`[Genus-CV]`, where no genus in a test fold appears in training; pooled out-of-fold metrics shown in table, 5-fold mean ± s.d. in text). Track 2 reveals 30% of each strain's assays to calibrate strain/EC posteriors and evaluates strictly on the held-out 70% assays. Because CarveMe, ModelSEED, and gapseq emit discrete 0/1 reaction calls rather than continuous probabilities, their ROC-AUC equals single-point balanced accuracy ($\frac{1}{2}$(TPR + TNR)), whereas GSPA and supervised classifiers emit continuous scores evaluated by threshold-swept ROC-AUC. Runtime reports per-genome execution after base reconstruction/feature extraction (end-to-end runtime including all 3 base models in parentheses).

Evaluation Track / Panel	Method / Predictor	Assays (n)	Precision	Recall	F_1 Score	Balanced Acc.	ROC-AUC	Runtime / Genome
Track 1 Blind: Carbon Growth (529 Strains)	CarveMe[54] [Blind]	2,332	0.6935	0.4073	0.5132	0.6233	0.6233 (bin.)	45.0 s
	ModelSEED[52] [Blind]	2,332	**0.8576**	0.5200	0.6474	0.7214	0.7214 (bin.)	180.0 s
	gapseq[55] [Blind]	2,332	0.8071	**0.7836**	**0.7952**	**0.8082**	0.8082 (bin.)	420.0 s
	GSPA Track 1 [100% Blind, Zero Labels]	2,332	0.8071	**0.7836**	**0.7952**	**0.8082**	**0.8567**	**6.1 s (651.1 s)**
Track 1 Supervised: Carbon Growth [Genus-CV]	BacDive RF Baseline[79,80] [Genus-CV]	2,332	0.8433	0.8464	0.8448	0.8530	0.9222	1.8 s
	DiASPora GBDT[80] [Genus-CV]	2,332	0.8475	0.8491	0.8483	0.8564	0.9268	1.8 s
	Traitar L1-SVM[81] [Genus-CV]	2,332	0.8472	**0.8673**	0.8571	0.8638	0.9185	1.8 s
	GSPA + DiASPora Hybrid [Genus-CV]	2,332	**0.8766**	0.8655	**0.8710**	**0.8783**	**0.9567**	**6.1 s (652.9 s)**
Track 1 Blind: Enzymes (3,017 Type Strains)	CarveMe[54] [Blind]	17,496	0.7739	0.2996	0.4319	0.5774	0.5774 (bin.)	45.0 s
	ModelSEED[52] [Blind]	17,496	0.7854	0.4779	0.5943	0.6310	0.6310 (bin.)	180.0 s
	gapseq[55] [Blind]	17,496	**0.8393**	**0.7204**	**0.7753**	**0.7461**	0.7461 (bin.)	420.0 s
	GSPA Track 1 [100% Blind, Zero Labels]	17,496	**0.8393**	**0.7204**	**0.7753**	**0.7461**	**0.7520**	**6.1 s (651.1 s)**
Track 1 Supervised: Enzymes [Genus-CV]	BacDive RF Baseline[79,80] [Genus-CV]	17,496	0.8391	0.8527	0.8458	0.7911	0.8593	1.8 s
	DiASPora GBDT[80] [Genus-CV]	17,496	0.8602	0.8499	0.8550	0.8108	0.8699	1.8 s
	Traitar L1-SVM[81] [Genus-CV]	17,496	0.8402	**0.8794**	0.8593	0.8014	0.8715	1.8 s
	GSPA + DiASPora Hybrid [Genus-CV]	17,496	**0.8652**	0.8776	**0.8713**	**0.8257**	**0.9097**	**6.1 s (652.9 s)**
Track 1: 10 Diagnostic ECs (6,313 Strains)	DeepProZyme vs. METEOR v6 / a025 / v7	50,075 (50,105)	0.6442 / 0.6194	0.8105 / 0.8640	0.7178 / 0.7215	0.6792 / 0.6639	0.6792 / 0.6639	18.0 / 313.8 s
Track 1: Urease `EC 3.5.1.5` (6,313 Strains)	DPZ vs. METEOR v6 (vs. **gapseq 17k**)	5,026 (871)	0.4520 / 0.4320	0.7891 / 0.8000	0.5748 / 0.5610 (**0.7626**)	0.7294 / 0.7184	0.7294 / 0.7184	18.0 / 313.8 s
Track 1: 108-Genome Catabolic (Glucose Mismatch)	DeepProZyme vs. METEOR v8 Minimal-Glucose	1,622	**0.6254** / 0.6051	0.7240 / **0.7559**	0.6711 / 0.6722	**0.5386** / 0.5101	**0.5716** / 0.5484	18.0 / 313.8 s
Track 2: Held-Out 70% Carbon (529 Strains)	CarveMe[54]	1,641	0.6927	0.3882	0.4975	0.6165	0.6165 (bin.)	45.0 s
	ModelSEED[52]	1,641	**0.8728**	0.5116	0.6451	0.7222	0.7222 (bin.)	180.0 s
	gapseq Track 1 (Unassisted)	1,641	0.8059	0.7789	0.7922	0.8049	0.8049 (bin.)	420.0 s
	GSPA Track 2 Bayesian Anchor (30% → 70%)	1,641	0.8362	**0.9447**	**0.8871**	**0.8889**	**0.9534**	**6.4 s (651.4 s)**
	Traitar L1-SVM[81] (30% → 70%)	1,641	0.8391	0.8380	0.8386	0.8466	0.9421	1.8 s
	BacDive RF Baseline[79,80] (30% → 70%)	1,641	0.8625	0.8625	0.8625	0.8692	0.9483	1.8 s
	DiASPora GBDT[80] (30% → 70%)	1,641	0.8727	0.8985	0.8854	0.8901	0.9575	1.8 s
	GSPA Track 2 + DiASPora Hybrid (30% → 70%)	1,641	**0.9048**	**0.9409**	**0.9225**	**0.9258**	**0.9788**	**6.4 s (653.2 s)**
Track 2: Held-Out 70% Enzymes (3,017 Strains)	CarveMe[54]	12,316	0.7791	0.3009	0.4341	0.5802	0.5802 (bin.)	45.0 s
	ModelSEED[52]	12,316	0.7868	0.4786	0.5952	0.6325	0.6325 (bin.)	180.0 s
	gapseq Track 1 (Unassisted)	12,316	**0.8376**	0.7188	0.7737	0.7446	0.7446 (bin.)	420.0 s
	GSPA Track 2 Bayesian Anchor (30% → 70%)	12,316	0.8067	**0.9084**	**0.8545**	**0.7749**	**0.8738**	**6.4 s (651.4 s)**
	BacDive RF Baseline[79,80] (30% → 70%)	12,316	0.8411	0.8575	0.8492	0.7953	0.8623	1.8 s
	DiASPora GBDT[80] (30% → 70%)	12,316	0.8541	0.8473	0.8507	0.8045	0.8691	1.8 s
	Traitar L1-SVM[81] (30% → 70%)	12,316	0.8301	**0.8854**	0.8569	0.7934	0.8728	1.8 s
	GSPA Track 2 + DiASPora Hybrid (30% → 70%)	12,316	**0.8638**	0.8819	**0.8727**	**0.8264**	**0.9080**	**6.4 s (653.2 s)**

(5-fold mean 0.8712 ± 0.0083, Precision = 0.8652, Recall = 0.8776, BAcc = 0.8257, ROC-AUC = 0.9097).

Third, in Track 2 (where 30% of each strain's assays are revealed to calibrate strain metabolic activity and co-occurring pathway posteriors, and evaluation is performed strictly on the held-out 70% assays), `GSPA Track 2 Bayesian Anchor` improves unsupervised held-out carbon-source F_1 (n = 1,641) from 0.7922 (gapseq and unassisted GSPA Track 1 on the same 70% split) to 0.8871 (Precision = 0.8362, Recall = 0.9447, BAcc = 0.8889, continuous ROC-AUC = 0.9534) and held-out enzyme F_1 (n = 12,316) from 0.7737 (gapseq) to 0.8545 (Precision = 0.8067, Recall = 0.9084, BAcc = 0.7749, continuous ROC-AUC = 0.8738). Combining Track 2 posteriors with DiASPora (`GSPA Track 2 + DiASPora Hybrid`) reaches F_1 = 0.9225 (Precision = 0.9048, Recall = 0.9409, BAcc = 0.9258, ROC-AUC = 0.9788) on held-out carbon sources and F_1 = 0.8727 (Precision = 0.8638, Recall = 0.8819, BAcc = 0.8264, ROC-AUC = 0.9080) on held-out diagnostic enzymes, outperforming DiASPora (0.8854 / 0.8507), BacDive RF (0.8625 / 0.8492), and Traitar (0.8386 / 0.8569; Table 4).

2.5 Obligate heteromeric protein complex completion (Suite 5)

We next evaluated completing obligate heteromeric protein complexes when individual subunits have twilight-zone homology while suppressing false-positive singleton subunit predictions (Suite 5; Figure 5a,b and Table 5; `results/pillar3_complex_lso.tsv`). Across curated heteromeric complexes from Complex Portal and GO (including proton-transporting ATP synthase `GO:0045259`, respiratory chain complex I `GO:0045271`, cytochrome bo_3/c oxidase, DNA-directed RNA polymerase, 30S/50S ribosomal subunits, and the SecYEG translocon), we performed 175 leave-subunit-out (LSO) trials across six prokaryotic proteomes (*E. coli* K-12 [n = 30], *B. subtilis* 168 [n = 30], *M. tuberculosis* H37Rv [n = 30], *P. aeruginosa* PAO1 [n = 30], *E. coli* O157:H7 [n = 30], and *Synechocystis* sp. PCC 6803 [n = 25]). In each trial, a true complex subunit's primary evidence was attenuated into the twilight zone ($q_e \in [0.22, 0.38]$) alongside five matched decoy proteins carrying comparable raw domain scores in unrelated genomic loci (6 candidates per trial; random-chance Hit@1 = 1/6 = 0.1667; Precision@0.5, Recall@0.5, and F_1@0.5 are evaluated per trial and macro-averaged across all trials).

Under baseline Noisy-OR pooling (`M1`), all 175 attenuated subunits remain below the θ = 0.50 thresh-

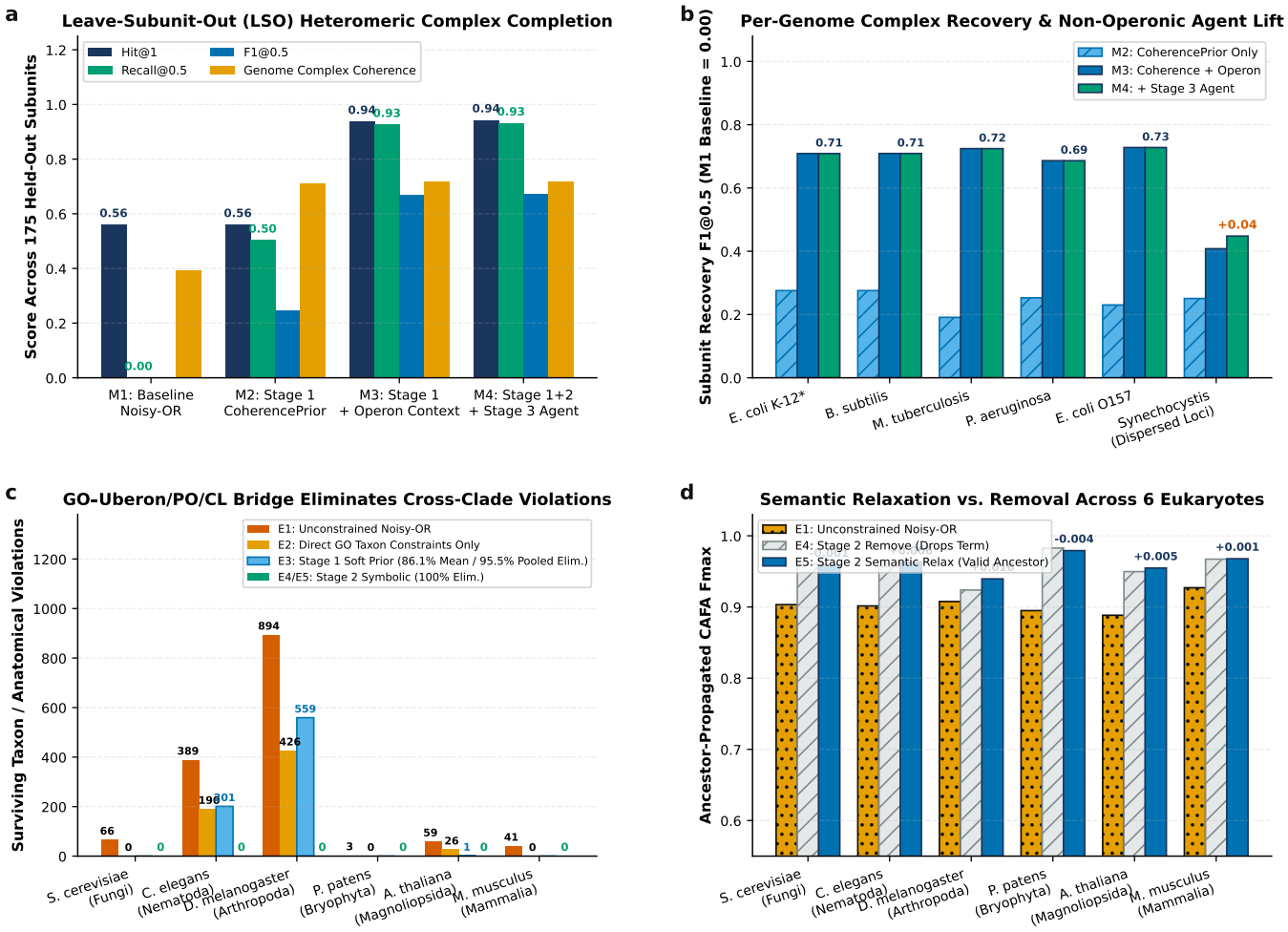


Figure 5. Obligate heteromeric protein complex completion and multicellular eukaryotic anatomical constraint enforcement (Suites 2 and 5). (**a**) Leave-subunit-out (LSO) recovery across 175 held-out heteromeric complex subunits in six prokaryotic genomes (chance Hit@1 = 0.167 over 6 candidates per case: 1 held-out subunit + 5 decoys), showing Hit@1, Recall@0.5, F_1@0.5, and Genome Complex Coherence across four configurations (`M1`–`M4`). (**b**) Per-genome subunit recovery F_1@0.5 across six prokaryotic species ($*$ marks the *E. coli* K-12 calibration reference), showing the +0.040 F_1 gain from the zero-leakage Stage 3 neuro-symbolic agent on *Synechocystis* sp. PCC 6803 where photosynthetic and respiratory complex subunits are dispersed across non-adjacent genomic loci. (**c**) Surviving cross-clade taxonomic and anatomical violations across six eukaryotic proteomes under unconstrained Noisy-OR (`E1`: 1,452 stress-test / 1,009 endogenous violations), direct GO taxon constraints only (`E2`: 55.8% stress-test elimination), Stage 1 soft anatomical prior (`E3`: 81.0% unweighted species mean / 47.6% pooled elimination), and Stage 2A deterministic enforcement (`E4`/`E5`: 100% elimination). (**d**) Ancestor-propagated transfer-evaluated micro $F_{\max}$ against unmodified Swiss-Prot ground truth across all six eukaryotes comparing unconstrained Noisy-OR (`E1`), Stage 2A naive removal (`E4`), and Stage 2A semantic ancestor relaxation (`E5`).

old (Recall@0.5 = 0.0000, F_1@0.5 = 0.0000, genome Complex Coherence = 0.3901), and Hit@1 among the 6 candidates is 0.5600 (95% CI $[0.483, 0.635]$). Enabling the Stage 1 complex coherence prior alone (`M2`) detects that the partner subunits of the complex are present in the genome and boosts all domain-matching candidates equally, leaving Hit@1 at 0.5600 while lifting Recall@0.5 to 0.5029, Precision@0.5 to 0.1729, and per-trial F_1@0.5 to 0.2459. Coupling complex coherence with the Stage 1 operon neighborhood prior (`M3`) disambiguates the true co-operonic subunit from non-operonic decoys: across all 175 trials, Hit@1 rises from 0.5600 to 0.9371 (95% CI $[0.890, 0.968]$), Recall@0.5 reaches 0.9257, Precision@0.5 rises more than three-fold to 0.5951, and F_1@0.5 reaches 0.6680 (0.9310 Hit@1, 0.9103 Recall@0.5, 0.5886 Precision@0.5, and 0.6595 F_1@0.5 across the 145 non-*E. coli* K-12 trials; Table 5).

Comparing per-genome recovery across the six species highlights the complementary roles of operon context and Stage 3 agentic reasoning (Figure 5b). In *E. coli* K-12, *B. subtilis*, *M. tuberculosis*, *P. aeruginosa*, and *E. coli* O157:H7, most heteromeric complex subunits reside in contiguous operons, allowing `M3` to achieve F_1@0.5 between 0.6862 (*P. aeruginosa*) and 0.7279 (*E. coli* O157:H7;

Table 5. Leave-subunit-out (LSO) recovery of 175 obligate heteromeric protein complex subunits across six prokaryotic reference genomes (Suite 5; source: `results/pillar3_complex_lso.tsv`). Each trial evaluates 1 true attenuated subunit ($q_e \in [0.22, 0.38]$) against 5 matched non-complex decoy proteins carrying comparable raw domain scores and symmetric zero-leakage formatting in unrelated genomic loci (6 candidates per case; random-chance Hit@1 = 1/6 = 0.1667; Precision@0.5, Recall@0.5, and F_1@0.5 are computed per trial and macro-averaged across trials). Middle rows report the 5-genome out-of-sample weighted mean ($n = 145$) excluding the hyperparameter development genome *E. coli* K-12.

Genome Scope	Method Configuration	Subunits (n)	Hit@1 (Chance = 0.167)	Recall@0.5	Precision@0.5	F_1@0.5	Complex Coh.
All 6 Genomes Combined	`M1`: Baseline Noisy-OR	175	0.5600	0.0000	0.0000	0.0000	0.3901
	`M2`: Stage 1 Complex Coherence Prior Only	175	0.5600	0.5029	0.1729	0.2459	0.7107
	`M3`: Stage 1 Coherence + Operon Context	175	0.9371	0.9257	0.5951	0.6680	**0.7170**
	`M4`: Stage 1 + Stage 2 + Stage 3 Agent	175	**0.9429**	**0.9314**	**0.6009**	**0.6737**	**0.7170**
5 Non-*E. coli* K-12 Genomes	`M3`: Stage 1 Coherence + Operon Context	145	0.9310	0.9103	0.5886	0.6595	**0.7011**
	`M4`: Stage 1 + Stage 2 + Stage 3 Agent	145	**0.9379**	**0.9172**	**0.5955**	**0.6664**	**0.7011**
Synechocystis sp. PCC 6803	`M1`: Baseline Noisy-OR	25	0.5200	0.0000	0.0000	0.0000	0.3587
	`M2`: Stage 1 Complex Coherence Prior Only	25	0.5200	0.4800	0.1813	0.2507	**0.6125**
	`M3`: Stage 1 Coherence + Operon Context	25	0.6400	0.6400	0.3393	0.4080	**0.6125**
	`M4`: Stage 1 + Stage 2 + Stage 3 Agent	25	**0.6800**	**0.6800**	**0.3793**	**0.4480**	**0.6125**

0.7089 in *E. coli* K-12, 0.7090 in *B. subtilis*, and 0.7244 in *M. tuberculosis*). In the cyanobacterium *Synechocystis* sp. PCC 6803, subunits of photosystem I/II, phycobilisomes, and NADH dehydrogenase are frequently split across dispersed chromosomal loci: baseline `M1` achieves 0.5200 Hit@1 (Complex Coherence = 0.3587), `M2` reaches 0.4800 Recall@0.5 and 0.2507 F_1@0.5 (Complex Coherence = 0.6125), and operon context (`M3`) reaches 0.6400 Hit@1 and 0.4080 F_1@0.5. Activating Stage 2 singleton pruning and the Stage 3 neuro-symbolic agent (`M4`) under symmetric zero-leakage candidate formatting elevates *Synechocystis* Hit@1 from 0.6400 to 0.6800 (17/25), Recall@0.5 from 0.6400 to 0.6800, Precision@0.5 from 0.3393 to 0.3793, and F_1@0.5 from 0.4080 to 0.4480 (+0.0400 gain), bringing overall 6-genome subunit recovery to 0.9429 Hit@1 (95% CI $[0.898, 0.972]$), 0.9314 Recall@0.5, 0.6009 Precision@0.5, and 0.6737 F_1@0.5 (0.9379 Hit@1, 0.9172 Recall@0.5, 0.5955 Precision@0.5, and 0.6664 F_1@0.5 on the 5 non-*E. coli* K-12 genomes).

2.6 Multicellular eukaryotic anatomical and developmental constraints (Suite 2)

We next evaluated anatomical and developmental constraint enforcement across six eukaryotic species spanning unicellular fungi (*Saccharomyces cerevisiae*), nonchordate invertebrates (*Caenorhabditis elegans* and *Drosophila melanogaster*), non-vascular bryophyte mosses (*Physcomitrium patens*), vascular flowering plants (*Arabidopsis thaliana*), and vertebrates (*Mus musculus*), evaluating both (i) the unperturbed endogenous homology-transfer proteome (6,155 curated proteins across the 6 species; `results/pillar3_eukaryote_endogenous_vs_injected.tsv`) and (ii) a cross-clade perturbation stress-test adding 1,546 cross-clade developmental claims across 7,655 proteins (300 in 5 species and 46 in *P. patens*), scored strictly against unmodified Swiss-Prot ground-truth annotations (Figure 5c,d and Table 6; `results/pillar3_eukaryotic_anatomy.tsv` and `results/pillar3_relax_vs_remove.tsv`). On the unperturbed endogenous proteomes alone, unconstrained Noisy-OR (`E1`) already produces 1,009 formal cross-clade anatomical and taxonomic violations transferred from mammalian or angiosperm homologs (3 in *S. cerevisiae*, 289 in *C. elegans*, and 717 in *D. melanogaster*; 0 in *P. patens*, *A. thaliana*, and *M. musculus*; mean 168.2 violations/species); GSPA eliminates 100% (1,009/1,009) of these endogenous violations and increases endogenous 6-species mean CAFA $F_{\max}$ from 0.7096 (`E1`) to 0.7117 at Iter 1 (+0.0021 mean gain; 0.7109 at Iter 3 with Stage 2A relaxation: +0.0086 in *S. cerevisiae* [0.6644 → 0.6730], +0.0032 in *C. elegans* [0.5797 → 0.5829], +0.0010 in *D. melanogaster* [0.5863 → 0.5873], +0.0010 in *P. patens* [0.8841 → 0.8851], and +0.0011 in *A. thaliana* [0.6594 → 0.6605]). Across the combined stress-test panel (1,452 total baseline violations at $\theta = 0.50$: 66 in *S. cerevisiae*, 389 in *C. elegans*, 894 in *D. melanogaster*, 3 in *P. patens*, 59 in *A. thaliana*, and 41 in *M. musculus*), applying direct GO taxon constraints alone (`E2`, using the 26,911 rules in `go-taxon-constraints.tsv`) eliminates 810/1,452 (55.79%) of violations while leaving 642/1,452 (44.21%) unresolved (190 in *C. elegans*, 426 in *D. melanogaster*, and 26 in *A. thaliana*), because those errors involve vertebrate, arthropod, or angiosperm organ terms whose taxonomic restrictions are defined in Uberon, PO, or CL rather than directly in GO. Applying the Stage 1 soft anatomical prior alone (`E3`, $-\alpha_4 = -4.0$ log-odds shift) eliminates 100% of violations in three species and 98.3% (58/59) in *A. thaliana*, but leaves 201 high-confidence violations in *C. elegans* (48.3% eliminated) and 559 in *D. melanogaster* (37.5% eliminated; 80.97% unweighted species mean elimination across the 6 species, or 47.59% [691/1,452] pooled), where raw homology scores exceed $q_e \geq 0.99$, demonstrating why Stage 2A deterministic enforcement (`E4` and `E5`, 0 surviving violations) is necessary after soft Bayesian updates.

Comparing unconstrained Noisy-OR (`E1`), Stage 2A naive removal (`E4`), and Stage 2A semantic ancestor relaxation (`E5`) on transfer-evaluated micro $F_{\max}$

Table 6. Anatomical and developmental constraint enforcement, semantic ancestor relaxation, and active-structure developmental rescue across six diverse eukaryotes under both cross-clade perturbation stress-test (7,655 proteins) and unperturbed endogenous-only (6,155 proteins) regimes evaluated against unmodified Swiss-Prot ground truth (Suite 2; source: `results/pillar3_eukaryotic_anatomy.tsv`, `results/pillar3_relax_vs_remove.tsv`, and `results/pillar3_eukaryote_endogenous_vs_injected.tsv`). Violation columns report surviving formal violations under unconstrained Noisy-OR (`E1`: stress-test violations with unperturbed endogenous violations in parentheses), direct GO taxon constraints only (`E2`), Stage 1 soft anatomical log-odds prior (`E3`), and Stage 2A deterministic enforcement (`E4` remove / `E5` relax). Ancestor-propagated transfer-evaluated micro $F_{\max}$, Recall, and Mean Information Content (IC, exact bits from `pillar3_relax_vs_remove.tsv`) compare `E1`, `E4`, and `E5`; bottom rows report the 6-species stress-test mean and the unperturbed endogenous-only 6-species CAFA $F_{\max}$ mean.

Species (Clade)	Taxon ID	E1 Viol. (Endog.)	E2 Direct TC Viol.	E3 Stage 1 Viol.	E4/E5 Viol.	E1 Noisy-OR $F_{\max}$ (Rec.)	E4 Remove $F_{\max}$ (Rec.)	E5 Relax $F_{\max}$ (Rec.)	$\Delta F_{\max}$ E5–E1 (E5–E4)	E1 / E4 / E5 Mean IC	Dev. Boost
S. cerevisiae (Fungi)	NCBITaxon_4932	66 (3)	0 (100.0%)	0 (100.0%)	**0** (100%)	0.9033 (0.9966)	**0.9604** (0.9249)	0.9592 (**0.9349**)	+0.0559 (−0.0012)	12.48 / 12.02 / **12.07**	2/3 (66.7%)
C. elegans (Nematoda)	NCBITaxon_6239	389 (289)	190 (51.2%)	201 (48.3%)	**0** (100%)	0.9015 (0.9720)	0.9567 (0.9268)	**0.9628 (0.9476)**	+0.0613 (+0.0061)	12.52 / 12.15 / 12.09	44/66 (66.7%)
D. melanogaster (Arthropoda)	NCBITaxon_7227	894 (717)	426 (52.3%)	559 (37.5%)	**0** (100%)	0.9076 (0.9378)	0.9238 (0.8608)	**0.9395 (0.8913)**	+0.0319 (+0.0157)	12.86 / 12.55 / 12.35	140/198 (70.7%)
P. patens (Bryophyta)	NCBITaxon_3218	3 (0)	0 (100.0%)	0 (100.0%)	**0** (100%)	0.8949 (0.9986)	**0.9828** (0.9697)	0.9792 (**0.9796**)	+0.0843 (−0.0036)	10.10 / 9.64 / **9.79**	0/0 (none in [0.22, 0.49])
A. thaliana (Magnoliopsida)	NCBITaxon_3702	59 (0)	26 (55.9%)	1 (98.3%)	**0** (100%)	0.8884 (0.9699)	0.9497 (0.9054)	**0.9547 (0.9232)**	+0.0663 (+0.0050)	11.76 / 11.13 / **11.18**	62/69 (89.9%)
M. musculus (Mammalia)	NCBITaxon_10090	41 (0)	0 (100.0%)	0 (100.0%)	**0** (100%)	0.9271 (0.9534)	0.9672 (0.9478)	**0.9678 (0.9545)**	+0.0407 (+0.0006)	13.01 / 12.84 / **12.85**	254/282 (90.1%)
Stress-Test 6-Eukaryote Mean (7,655 proteins; 1,452 viol.)		242.0 (1,009 endog.)	107.0 (55.8%)	126.8 (81.0% sp.)	**0** (100%)	0.9038 (0.9714)	0.9568 (0.9226)	**0.9605 (0.9385)**	+0.0567 (+0.0037)	12.12 / 11.72 / **11.72**	502/618 (81.2%)
Unperturbed Endogenous-Only 6-Eukaryote Mean (6,155 proteins)		168.2 (1,009 tot.)	102.7 (38.9%)	126.7 (24.7%)	**0** (100%)	0.7096 (CAFA)	0.7109 (Iter 3)	**0.7117 (Iter 1)**	+0.0021 (+0.0013 I3)	—	—

against unmodified Swiss-Prot ground truth clarifies the exact regime in which ancestor relaxation outperforms naive deletion (Figure 5d and Table 6; `results/pillar3_relax_vs_remove.tsv`). Naive deletion (`E4`) strips ancestral GO branches whenever an overly specific cross-clade developmental term is transferred, causing recall to drop by 0.56 to 7.70 percentage points across the six species. Under semantic ancestor relaxation (`E5`), forbidden terms are relaxed to their nearest anatomically valid GO ancestor, retaining 96.0%–98.8% of unconstrained `E1` mean annotation Information Content (12.07 vs. 12.48 bits in *S. cerevisiae*, 12.09 vs. 12.52 in *C. elegans*, 12.35 vs. 12.86 in *D. melanogaster*, 9.79 vs. 10.10 in *P. patens*, 11.18 vs. 11.76 in *A. thaliana*, and 12.85 vs. 13.01 in *M. musculus*; Table 6). Across all four multicellular animals and vascular plants (*C. elegans*, *D. melanogaster*, *A. thaliana*, and *M. musculus*), where homologous developmental regulators genuinely participate in ancestral tissue and organ morphogenesis, `E5` outperforms `E4` in both micro $F_{\max}$ ($\Delta F_{\max}(\text{E5}-\text{E4}) = +0.0006$ in *M. musculus*, +0.0050 in *A. thaliana*, +0.0061 in *C. elegans*, and +0.0157 in *D. melanogaster*; 6-species mean $F_{\max}$ rising from 0.9038 [`E1`] and 0.9568 [`E4`] to 0.9605 [`E5`]) and Recall (+0.67 to +3.05 percentage points). Conversely, in unicellular yeast (*S. cerevisiae*) and sparsely annotated non-vascular moss (*P. patens*), where injected organ-development claims land on metabolic or housekeeping proteins whose unmodified Swiss-Prot annotations lack multicellular developmental ancestors, `E5` still recovers higher Recall (+1.00 and +0.99 percentage points) and higher Mean IC (12.07 vs. 12.02 and 9.79 vs. 9.64 bits) than `E4`, whereas complete removal (`E4`) achieves slightly higher precision and $F_{\max}$ (0.9604 vs. 0.9592 in yeast; 0.9828 vs. 0.9792 in moss). In addition, for true species-appropriate developmental annotations attenuated into the borderline regime ($P^{(0)} \in [0.22, 0.49]$), the active-structure developmental boost in Stage 1 (`E3`) rescues 502/618 (81.2%) of borderline developmental claims above $\theta = 0.50$.

2.7 Evaluation of the Stage 3 neuro-symbolic agentic layer and extremophile case study (Suite 8)

Finally, we evaluated the optional Stage 3 neuro-symbolic agentic completion layer (Suite 8) against deterministic Stage 1+2, a posterior threshold baseline ($\geq$ 0.35 + Guardrail), and supervised non-agentic tabular classifiers (Logistic Regression, Random Forest, Gradient-Boosted Decision Trees, and Multi-Layer Perceptron trained via 5-fold cross-validation; Figure 6 and Table 7; `results/pillar4_agent_evaluation.tsv`, `results/pillar4_ml_baselines.tsv`, and `results/pillar4_multiseed_summary.tsv`). Across the 88-proteome multi-kingdom benchmark, the Stage 3 anomaly triggers select a mean of 1.84% of proteins per proteome (interquartile range 1.12%–2.38%, $< 2.5\%$ across all four domains). To evaluate anomaly resolution under strictly controlled, de-identified conditions without gene-symbol or formatting leakage, we constructed a 120-anomaly benchmark (30 pathway gaps, 30 incomplete complexes, 30 borderline essentiality gaps, and 30 borderline conflicts across 7 genomes: *E. coli* K-12 [$n = 29$], *P. aeruginosa* [$n = 18$], *B. subtilis* [$n = 17$], *M. tuberculosis* [$n = 16$], *S. cerevisiae* [$n = 16$], *C. elegans* [$n = 13$], and *A. thaliana* [$n = 11$]; `results/pillar4_agent_benchmark_120.jsonl`). Every candidate protein in the benchmark has its gene symbol masked to `"uncharacterized"`, candidate order is alternated across anomalies, and all true-positive and decoy candidates expose symmetric domain and real co-strand proteome neighbor feature formatting. The 90 completeness anomalies comprise 54 true-positive promotable gaps (18 per category) and 36 matched-score true-negative decoys (12 per category) including hard feature-overlap cases where decoys share borderline posteriors (0.24–0.36) or moderate RLGC Bayes factors (1.5–4.2) with true gaps, while the 30 `BORDERLINE_CONFLICT` anomalies serve as a symbolic-verification unit test comprising 15 real cross-clade conflicts that require ancestor relaxation and 15

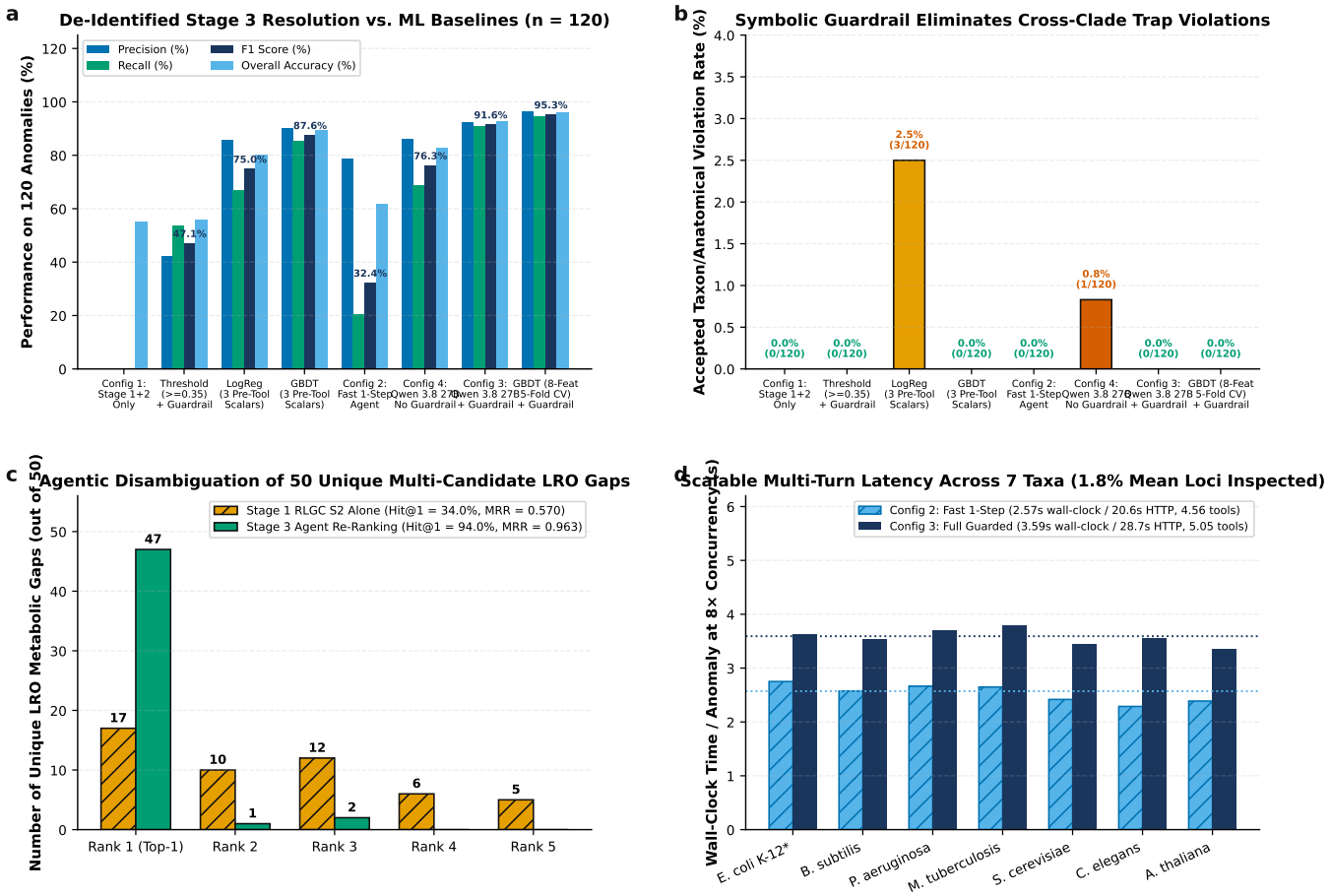


Figure 6. Evaluation and ablation of the optional Stage 3 neuro-symbolic agentic completion layer on the de-identified 120-anomaly and 50-unique-protein LRO benchmarks (Suite 8). (**a**) Precision, Recall, F_1 score, and overall resolution accuracy across 120 de-identified post-Stage-2 genome-scale anomalies (30 pathway gaps, 30 incomplete complexes, 30 borderline essentiality gaps, and 30 borderline conflicts) comparing deterministic Stage 1+2 only (Config 1), posterior threshold ≥ 0.35 + Guardrail, 3-scalar and 8-feature supervised 5-fold CV tabular baselines (LogReg, RF, GBDT, MLP), Fast 1-Step single-turn mode (Config 2), `Qwen/Qwen3.8-27B-FP8` without the symbolic guardrail (Config 4), and full multi-turn `Qwen/Qwen3.8-27B-FP8` with the symbolic ontology guardrail (Config 3, single seed and 3-seed mean ± s.d.). (**b**) Accepted taxonomic and anatomical violation rate across the 120 anomalies, showing that unguarded tabular classifiers and unguarded LLMs accept 0.83%–2.50% violations whereas the symbolic guardrail guarantees 0.0% (0/120). (**c**) Rank distribution of the true enzyme across 50 unique ambiguous multi-candidate LRO metabolic gaps before and after Stage 3 agentic re-ranking (Hit@1 rises from 34.0% [17/50] to 94.0% [47/50] at seed 42 and 96.7% ± 1.9% across 3 seeds, MRR from 0.5700 to 0.9633 / 0.9789). (**d**) Mean concurrency-normalized wall-clock time per anomaly (3.59 s at 8× async concurrency; 28.74 s single-request HTTP latency) across seven prokaryotic and eukaryotic species.

adversarial cross-clade traps that must be blocked.

Deterministic Stage 1+2 alone (Config 1) leaves post-Stage-2 borderline gaps unpromoted (0/54 promotions, accuracy = 55.00%), while simply lowering the posterior decision threshold to 0.35 with the symbolic guardrail (`Thresh-0.35`) promotes 29 true gaps alongside 40 false positives (Precision = 0.4203, Recall = 0.5370, F_1 = 0.4715). Supervised tabular classifiers trained via 5-fold CV on the 3 pre-tool summary scalars (`[current_posterior, diamond_score, suggestion_score]`) achieve F_1 = 0.7500 (Precision = 0.8571, Recall = 0.6667, 3/120 [2.5%] violations accepted) for Logistic Regression (`CV-LR3`) and F_1 = 0.8762 (Precision = 0.9020, Recall = 0.8519, Accuracy = 0.8917) for GBDT (`CV-GB3`), while reaching 96.0% (MRR = 0.9750) and 98.0% (MRR = 0.9850) Hit@1 on the 50 ambiguous LRO cases when trained via 5-fold CV on those cases' 3 pre-tool scalars. When supervised 5-fold CV classifiers are trained directly on all 8 pre-extracted oracle tool features (`[posterior, diamond, rlgc_bf, in_operon, pfam_count, partner_count, coverage, num_candidates]`) and paired with the deterministic `SymbolicGuardrail` (`CV-LR8+G`, `CV-RF8+G`, `CV-GB8+G`, and `CV-MLP8+G`), they establish a strong in-distribution supervised upper bound on binary promotion

Table 7. Ablation of the optional Stage 3 neuro-symbolic agent (`Qwen/Qwen3.8-27B-FP8`) against deterministic, threshold, and supervised 5-fold CV tabular baselines across 120 de-identified post-Stage-2 genome-scale anomalies and 50 ambiguous LRO metabolic gaps (Suite 8; source: `results/pillar4_agent_evaluation.tsv`, `results/pillar4_ml_baselines.tsv`, and `results/pillar4_multiseed_summary.tsv`). All candidate gene symbols are masked to `"uncharacterized"`, candidate order is alternated, and domain/neighbor strings expose symmetric zero-leakage InterPro/Pfam and GO/EC identifiers with real proteome accessions. The 120 anomalies (30 per category across 7 genomes) comprise 54 true-positive promotable gaps (18 per completeness category), 36 matched-score true-negative decoys (12 per category), and 30 borderline conflict cases (15 real cross-clade conflicts and 15 adversarial cross-clade traps; overall resolution accuracy credits any non-promotion [`RELAX`, `DEMOTE`, or `ABSTAIN`] on real conflicts, while exact ancestral GO relaxation counts are reported under **TP Prom. / Relax**). Latency reports concurrency-normalized wall-clock seconds per anomaly at 8× async concurrency (with mean per-request HTTP latency in parentheses).

Config	Stage 3 Architecture / Baseline	TP Prom. / Relax	Precision	Recall	F_1 Score	Accuracy	Viol. Accepted	LRO-50 Hit@1 (MRR)	Tool Calls	Wall-Clock (HTTP)
Config 1	Stage 1 + Stage 2 Deterministic Only	0 / 0	0.0000	0.0000	0.0000	0.5500	**0 / 120 (0.0%)**	34.0% (0.5700)	0.00	0.00 s (0.00 s)
Thresh-0.35	Posterior Threshold (≥ 0.35) + Symbolic Guardrail	29 / 11	0.4203	0.5370	0.4715	0.5583	**0 / 120 (0.0%)**	34.0% (0.5700)	0.00	< 0.01 s
CV-LR3	Logistic Regression (5-fold CV, 3 Pre-Tool Scalars, No Guard)	36 / 0	0.8571	0.6667	0.7500	0.8000	3 / 120 (2.5%)	96.0% (0.9750)	0.00	< 0.01 s
CV-GB3	GBDT (5-fold CV, 3 Pre-Tool Scalars, No Guard)	46 / 0	0.9020	0.8519	0.8762	0.8917	**0 / 120 (0.0%)**	98.0% (0.9850)	0.00	< 0.01 s
Config 2	Fast 1-Step Agent (`-max-turns 2`, Guardrail ON)	11 / 3	0.7857	0.2037	0.3235	0.6167	**0 / 120 (0.0%)**	40.0% (0.6133)	4.56	2.57 s (20.57 s)
Config 4	`Qwen/Qwen3.8-27B-FP8` without Guardrail	37 / 0	0.8605	0.6852	0.7629	0.8250	1 / 120 (0.83%)	86.0% (0.9167)	4.38	4.13 s (33.01 s)
Config 3	**`Qwen/Qwen3.8-27B-FP8` + Symbolic Guardrail (Seed 42)**	**49 / 9**	0.9245	0.9074	0.9159	0.9250	**0 / 120 (0.0%)**	**94.0% (0.9633)**	5.05	3.59 s (28.74 s)
Config 3 (3-Seed)	**`Qwen/Qwen3.8-27B-FP8` + Guardrail (3-Seed Mean ± s.d.)**	**49.0 / 9.7**	**0.9506 ± 0.0349**	**0.9074 ± 0.0302**	**0.9275 ± 0.0084**	**0.9389 ± 0.0104**	**0 / 120 (0.0%)**	**96.7 ± 1.9% (0.9789)**	5.05	3.59 s (28.74 s)
CV-LR8+G	Logistic Regression (5-fold CV, 8 Oracle-Tool Feat. + Guardrail)	49 / 1	0.9800	0.9074	0.9423	0.9500	**0 / 120 (0.0%)**	**100.0% (1.0000)**	0.00	< 0.01 s
CV-RF8+G	Random Forest (5-fold CV, 8 Oracle-Tool Feat. + Guardrail)	52 / 0	0.9286	0.9630	0.9455	0.9500	**0 / 120 (0.0%)**	96.0% (0.9733)	0.00	< 0.01 s
CV-GB8+G	GBDT (5-fold CV, 8 Oracle-Tool Feat. + Guardrail)	51 / 0	0.9623	0.9444	0.9533	0.9583	**0 / 120 (0.0%)**	94.0% (0.9600)	0.00	< 0.01 s
CV-MLP8+G	**MLP (5-fold CV, 8 Oracle-Tool Feat. + Guardrail, In-Dist. Upper)**	52 / 0	**0.9811**	**0.9630**	**0.9720**	**0.9750**	**0 / 120 (0.0%)**	98.0% (0.9900)	0.00	< 0.01 s

(F_1 = 0.9423–0.9720, Accuracy = 0.9500–0.9750, and LRO-50 Hit@1 = 94.0%–100.0% in < 0.01 s). However, these supervised classifiers require labeled anomaly training splits from the benchmark distribution and cannot propose semantic ancestor GO terms on the 15 relaxable conflict cases (0–1 relaxations).

Operating zero-shot without any anomaly training labels across a bounded multi-turn loop (≤ 6 turns, mean 5.05 tool calls, 28.74 s single-request HTTP latency or 3.59 s wall-clock time per anomaly at 8× asynchronous concurrency) with the symbolic ontology guardrail active (Config 3, `Qwen/Qwen3.8-27B-FP8`), the Stage 3 agent achieves single-seed (seed 42) Precision = 92.45% (49/53), Recall = 90.74% (49/54), F_1 = 0.9159, and overall resolution accuracy = 92.50% (111/120, where any non-promotion [`RELAX`, `DEMOTE`, or `ABSTAIN`] on the 15 real conflicts is credited as avoiding a false promotion: 49/54 TP promotions, 32/36 decoy abstentions, 9/15 exact ancestral GO relaxations + 6/15 demotions on real conflicts, and 14/15 cross-clade traps rejected directly by the LLM with the 1 remaining raw trap proposal blocked by `SymbolicGuardrail` for 0/120 = 0.0% accepted ontology violations), and across 3 independent random seeds (seed $\in \{42, 43, 44\}$; `results/pillar4_multiseed_summary.tsv`) achieves Precision = 0.9506 ± 0.0349, Recall = 0.9074 ± 0.0302, F_1 = 0.9275 ± 0.0084, Accuracy = 0.9389 ± 0.0104, 9.7 ± 0.6 valid ancestral GO relaxations (9 at seed 42, 10 at seeds 43 and 44), and 0.00 ± 0.00% accepted ontology violation rate (Table 7). Zero-shot Config 3 outperforms the pre-tool 3-scalar CV classifiers on the 120-anomaly benchmark ($F_1 = 0.9275$ vs. 0.7500–0.8762) and approaches the in-distribution 8-feature supervised CV upper bound ($F_1 = 0.9423$–0.9720) without requiring training labels. In Fast 1-Step mode (Config 2, ≤ 2 turns, 4.56 tool calls, 2.57 s wall-clock / 20.57 s HTTP latency), precision is 0.7857 (11/14) but recall drops to 0.2037 ($F_1 = 0.3235$). Ablating the symbolic guardrail (Config 4, 4.38 tool calls, 4.13 s wall-clock / 33.01 s HTTP latency) allows 1/120 (0.83%) cross-clade traps to be accepted and reduces recall on relaxable conflicts, lowering precision to 0.8605, recall to 0.6852, F_1 to 0.7629, and accuracy to 0.8250 (Figure 6b). On the 50 deduplicated unique-protein ambiguous LRO metabolic gaps in *E. coli* K-12 (`results/pillar4_lro_reranking.tsv`), where all gene symbols are masked to `"uncharacterized"`, candidate ordering is randomized, and all 5 candidates per case expose symmetric domain and real proteome operon-neighbor metadata, zero-shot Stage 3 agentic re-ranking elevates Hit@1 from 34.0% (17/50, RLGC baseline) to 94.0% (47/50 at seed 42; two-sided exact binomial McNemar $p = 1.86 \times 10^{-9}$ on $b_{01} = 30, b_{10} = 0$ discordant unique proteins) and $96.67\% \pm 1.89\%$ across 3 seeds (98.0% [49/50] at seeds 43 and 44), Hit@3 from 86.0% (43/50) to 100.0% (50/50), and MRR from 0.5700 to 0.9633 (seed 42) and 0.9789 ± 0.0110 (3-seed mean; Figure 6c and Table 7), matching the 5-fold CV supervised tabular models (94.0%–100.0%) without using any LRO training folds. Finally, as an exploratory ecological case study on five desert extremophile isolates from the Empty Quarter (*Rub' al Khali*: *Salinicola* sp. [3,842 CDS], *Halomonas* sp. [4,118 CDS], *Pseudomonas* sp. [5,490 CDS], *Bacillus endophyticus* [4,625 CDS], and *Streptomyces* sp. [7,184 CDS]; `results/pillar6_empty_quarter_vignette.tsv`), applying GSPA lifts mean core essential completeness from 0.8884 (raw homology) and 0.9190 (Stage 0) to 0.9718, increases pathway coherence from 0.4892 to 0.6226 (+13.34 percentage points), and identifies 4–8 (mean 5.6) flux-certified pFBA core repairs and 4–9 (mean 6.2) Stage 3 candidate promotions per isolate across ectoine/betaine osmoprotection and oxidative stress pathways.

3 Discussion

Protein function prediction has traditionally been formulated as a per-protein classification problem evaluated on isolated sequences or structures[1,21,24]. Our evaluation across eight benchmark suites demonstrates that while per-protein multi-channel calibration (Stage 0) establishes bulk protein-centric accuracy on well-annotated reference proteomes, treating the whole genome as the unit of inference (Stages 1–3) resolves structural failure modes that per-protein predictors and negative-cost MILP gap-fillers cannot address.

Four main conclusions emerge from our study. First, our audit of 108 bacterial metabolic reconstructions clarifies why negative-cost MILP gap-filling (including our prior METEOR formulation[61]) degrades gene-level enzyme precision on diagnostic and catabolic panels: assigning negative addition costs ($c_j^{\text{add}} < 0$) to candidate reactions above an addition threshold causes 68.62% (215.02/313.35) of added reactions to carry zero steady-state flux, while uniform reaction-to-gene boosting inflates false-positive calls across paralogs. While any positive-cost LP formulation over the sanitized universal network $\mathcal{U}_{\text{clean}}$ eliminates uncertified zero-flux additions (81.31/81.31 flux-certified additions under threshold LP repair and 5.70/5.70 under two-step lexicographic pFBA with FVA/KO/perturbation certification in 18.01 s/genome), combining role-aware DNF mechanism aggregation (Top-M_{iso} isozyme OR and bottleneck-gated geometric-mean subunit AND), two-step lexicographic pFBA repair, cognate substrate flux probing, and competitive locus responsibility against an explicit unassigned-gene hypothesis $h_{\varnothing}$ eliminates the 2.23–2.36 dead-end active reactions per genome left by unregularized LP/MILP repair, improves `Price-149` enzyme Top-1 hits under both single-predictor refinement (CLEAN 60 → 63, MAPred 50 → 56, DeepProZyme 49 → 51 with cognate SLR, GraphEC 18 → 20) and multi-predictor refinement (+0.0528 to +0.1987 $F_{\max}$), and improves `BacDive` phenotype predictions across blind, Track 2 phenotype-anchored, and genus-blocked evaluation tracks.

Second, our ablation of the leave-reaction-out (LRO) benchmark separates the contribution of protein language model embeddings from that of local genomic synteny. On the well-studied *E. coli* K-12 proteome (Track B), per-protein ESM-2 reaction-centroid similarity alone recovers 62.11% of masked reactions at Rank 1 (87.85% given the candidate pool), with Stage 1 pathway coherence and RLGC operon neighborhood density adding +1.56 percentage points unconditionally (+2.21 percentage points given pool, to 63.67% / 90.06%). Conversely, when close PLM donor centroids are masked (Track A, cosine > 0.50) or when multiple candidate genes in a proteome share similar domain and embedding scores ($n = 50$ unique-protein ambiguous LRO gaps and $n = 175$ heteromeric complex LSO trials), sequence similarity alone recovers only 1.95% (LRO), 34.00% (ambiguous LRO), and 56.00% (LSO), whereas RLGC operon neighborhood coupling, Stage 1 coherence, and Stage 3 neuro-symbolic disambiguation elevate Hit@1 to 23.44% (Track A LRO), 96.67% (3-seed mean on ambiguous LRO), and 94.29% (LSO across 6 candidates per trial).

Third, our multi-kingdom eukaryotic evaluation shows that enforcing ontology consistency in multicellular organisms requires bridging the Gene Ontology with external anatomy ontologies (Uberon, PO, and CL). On unperturbed 7-channel predictions across six model eukaryotes, GSPA eliminates all 1,009 endogenous homology-transferred violations while improving mean CAFA $F_{\max}$ by +0.0021 (+0.0086 in *S. cerevisiae*). Under cross-clade perturbation stress-testing evaluated against unmodified Swiss-Prot ground truth (1,452 total baseline violations across fungi, nematodes, insects, mosses, flowering plants, and mammals), direct GO taxon constraints leave 44.21% (642/1,452) of errors unresolved and soft Stage 1 log-odds penalties still leave 201–559 high-score violations in invertebrate proteomes, whereas Stage 2A semantic relaxation over our 2,715 GO–Uberon/PO/CL cross-product axioms eliminates 100% of violations, retains 96.0%–98.8% of unconstrained mean IC, and outperforms naive deletion across all four multicellular animals and vascular plants ($\Delta F_{\max}(\text{E5} - \text{E4}) = +0.0006$ to +0.0157; Recall +0.67 to +3.05 percentage points), while recovering higher Recall and Mean IC than deletion in unicellular yeast and non-vascular moss.

Fourth, comparing the zero-shot Stage 3 neuro-symbolic agent (`Qwen/Qwen3.8-27B-FP8`) against both pre-tool 3-scalar and oracle-tool 8-feature supervised 5-fold CV classifiers on the de-identified 120-anomaly and 50-LRO benchmarks clarifies when an LLM agent is warranted over a lightweight tabular classifier. When in-distribution labeled anomaly data and pre-extracted tool features are available and only binary promotion or candidate ranking is needed, an 8-feature supervised classifier paired with `SymbolicGuardrail` achieves $F_1 = 0.9423$–0.9720 on the 120 anomalies and 94.0%–100.0% Hit@1 on LRO-50 in < 0.01 s. However, when labeled anomaly training data are unavailable for a new clade or when multi-class semantic ancestor relaxation is required on cross-clade conflicts, zero-shot multi-turn tool use with `SymbolicGuardrail` achieves $F_1 = 0.9275 \pm 0.0084$ and accuracy $= 0.9389 \pm 0.0104$ on the 120 anomalies and 96.67% ± 1.89% Hit@1 on LRO-50 across 3 seeds, outperforming pre-tool 3-scalar CV classifiers on anomaly resolution ($F_1 = 0.7500$–0.8762), proposing 9.7/15 valid ancestral GO relaxations, and guaranteeing a 0.0% accepted ontology violation rate.

4 Methods

4.1 Overview of the four-stage architecture and notation

Let $\mathcal{P} = \{p_1, \ldots, p_N\}$ denote the protein-coding genes of a target genome with NCBI Taxonomy identifier τ_{tax}, genomic coordinates $\mathcal{K} = \{(c_i, s_i, e_i, \zeta_i)\}_{i=1}^{N}$ (contig c_i, start s_i, end e_i, and strand $\zeta_i \in \{+, -\}$), and let $\mathcal{T}$ denote the directed acyclic graph (DAG) of the Gene Ontology with true-path relation $t \sqsubseteq a$ indicating that term t is a subtype or part of ancestor a. GSPA estimates the proteome-wide annotation posterior over $\mathbf{Y} \in \{0,1\}^{N \times |\mathcal{T}|}$ conditioned on genomic coordinates $\mathcal{K}$, taxon τ_{tax}, metabolic network $\mathcal{M}$, and cross-ontology axioms $\mathcal{O}$ through four sequential stages (Figure 1; benchmark suite organization, Stage 0 channel configuration, and active stages per suite are summarized in Table 8).

4.2 Stage 0: Multi-channel per-protein evidence calibration and Noisy-OR pooling

GSPA can refine either a single standalone predictor (such as CLEAN[27], MAPred[30], DeepProZyme[28], EnzBERT[29], GraphEC[31], or TopEC[32]) or integrate claims from up to seven per-protein evidence channels $e \in \mathcal{E}_{\text{ch}}$: (1) sequence alignment via DIAMOND[12] against Swiss-Prot GOA entries; (2) profile HMM domain architecture via HMMER 3[13] against Pfam-A/Pfam2GO; (3) domain signatures via InterProScan v5[10]; (4) ESM-2 (650M) embedding kNN transfer[14]; (5) ProtT5-XL-U50 embedding kNN transfer[15]; (6) sequence-based neural classifiers (ProteInfer[25], DeepGOWeb[83], DeepGO-SE[24]); and (7) structural similarity transfer via FoldSeek[33] and ProstT5 3Di[16].

For each channel $e \in \mathcal{E}_{\text{ch}}$, raw scores are mapped to calibrated probabilities $\tilde{q}_e(p,t) \in (0,1)$ in `CalibrationTable.groovy` via piecewise-linear interpolation over pre-fitted isotonic calibration knots derived from a temporally pre-cutoff Swiss-Prot validation split (20% of pre-t_0 Swiss-Prot proteins, strictly disjoint from all Suite 1–8 test proteins; falling back to a parameterized logistic curve $\sigma(k(s - s_0))$ when custom knot tables are omitted) and multiplied by the GO evidence-code reliability weight $w_{\text{ev}}(c)$: experimental codes (`EXP, IDA, IPI, IMP, IGI, IEP, HTP, HDA, HMP, HGI, HEP`) receive $w_{\text{ev}} = 1.00$, curated computational/author statements (`ISS, ISO, ISA, ISM, IGC, RCA, TAS, NAS, IC`) receive $w_{\text{ev}} = 0.80$ (0.65 for `RCA/NAS` in `EvidenceType.groovy`), phylogenetic codes (`IBA, IBD, IKR, IRD`) receive $w_{\text{ev}} = 0.70$–0.75, and electronic annotations (`IEA`) receive $w_{\text{ev}} = 0.50$–0.55, yielding $q_e(p,t) = w_{\text{ev}}(c)\tilde{q}_e(p,t)$. To prevent correlated channels from over-counting redundant evidence, `EvidenceCombiner.groovy` partitions claims into correlation groups $\mathcal{G}_{\text{corr}}$ (defaulting to sequence-homology vs. domain-signature vs. structure/PLM groups, or donor-accession groups in the Python multi-channel runner where $n_{\text{don}}(p,t)$ channels sharing the same Swiss-Prot donor accession $d_e(p,t)$ receive discount exponent $\rho_e(p,t) = 1/\sqrt{n_{\text{don}}(p,t)}$; in `EvidenceCombiner.groovy`, redundant claims within the same correlation group $g \in \mathcal{G}_{\text{corr}}$ are collapsed to their group maximum $q_g(p,t) = \max_{e \in g} q_e(p,t)$ prior to across-group Noisy-OR pooling). Pooling across channels and upward along the GO DAG $\mathcal{T}$ gives:

$$P_0(p,f) = 1 - \prod_{t \sqsubseteq f} \prod_{e \in \mathcal{E}_{\text{ch}}(p,t)} \big(1 - q_e(p,t)\big)^{\rho_e(p,t)}. \quad (1)$$

Clamping $P_0(p,f)$ to $[\varepsilon, 1-\varepsilon]$ ($\varepsilon = 10^{-4}$ in Python; 10^{-6} in `IterativeRefiner.groovy`) yields the anchored Stage 0 log-odds $L^{(0)}(p,f) = \log \frac{P_0(p,f)}{1-P_0(p,f)}$.

4.3 Stage 1: Proteome-wide iterative Bayesian log-odds refinement

In Stage 1 (`IterativeRefiner.groovy`), GSPA updates the log-odds matrix $\mathbf{L}^{(k)}$ across iterations $k = 1, \ldots, K_{\max}$ by adding six genome-scale prior shifts to the anchored Stage 0 log-odds $L^{(0)}(p,f)$ as a function of the previous iterate's probabilities $P^{(k-1)}(p,f) = \sigma_{\log}\big(L^{(k-1)}(p,f)\big)$, where $\sigma_{\log}(x) = (1+e^{-x})^{-1}$, with damping factor $\gamma_{\text{iter}} = 0.50$:

$$\begin{aligned} \tilde{L}^{(k)}(p,f) &= L^{(0)}(p,f) \\ &\quad + \sum_{m=1}^{6} \alpha_m \, B_m\big(p, f \mid \mathbf{P}^{(k-1)}, \mathcal{K}, \tau_{\text{tax}}, \mathcal{M}, \mathcal{O}\big), \\ P^{(k)}(p,f) &= (1-\gamma_{\text{iter}}) P^{(k-1)}(p,f) + \gamma_{\text{iter}} \sigma_{\log}\big(\tilde{L}^{(k)}(p,f)\big). \end{aligned} \quad (2)$$

After each iteration, updated probabilities are propagated upward on $\mathcal{T}$ via $P^{(k)}(p,a) \leftarrow \max_{t \sqsubseteq a} P^{(k)}(p,t)$ (and whenever a parent term a is suppressed below 0.50 by a taxonomic or anatomical penalty, its descendants $t \sqsubseteq a$ are capped at $\min(P^{(k)}(p,t), P^{(k)}(p,a))$). In `IterativeRefiner.groovy`, the inner loop runs for up to $K_{\max} = 6$ iterations with convergence tolerance 5×10^{-3} on the mean probability change across active pairs and automatic divergence rollback if the mean change increases across consecutive iterations (in the standalone Python LRO/EC runners, up to $K_{\max} = 20$ undamped iterations with $\max_{p,f} |P^{(k)}(p,f) - P^{(k-1)}(p,f)| < 10^{-3}$). The six functional forms $B_1, \ldots, B_6$ are implemented as follows:

1. **Essentiality prior ($m = 1$, $\alpha_1 = 1.5$, `EssentialityPrior.groovy`):** Let $\mathcal{F}_{\text{ess}}(\tau_{\text{tax}})$ be the domain-specific essential function profile and $P^{(k-1)}_{\max}(f) = \max_{p \in \mathcal{P}} P^{(k-1)}(p,f)$. For $f \in \mathcal{F}_{\text{ess}}(\tau_{\text{tax}})$ with $P^{(k-1)}_{\max}(f) < \theta_{\text{ess}} = 0.50$, let $\text{rank}_f(p) \in \{1, 2, \ldots\}$ be the rank of protein p among candidates with $P^{(k-1)}(p,f) \geq 0.05$ ordered by descending $P^{(k-1)}(p,f)$ (ties broken deterministically by descending Stage 0 score $P_0(p,f)$ and locus identifier); then

Table 8. Overview of the eight benchmark suites, nomenclature mapping, Stage 0 channel configuration, active GSPA stages, and canonical data files in `results/`. Full column-level provenance is documented in `results/PROVENANCE_MAP.md`.

Suite	Internal Pillar / Scenario	Cohort & Scale	Stage 0 Channels	Active GSPA Stages	Primary Comparators	Canonical results/ File(s)
Suite 1	Pillar 1 / Scen. 1A–1C	CAFA holdout (16,125), CAFA6 (769), EC-125, 95 genomes (69,666)	7-ch GO / 3-pred EC / 1-pred EC	Stage 0 + 1 + 2A (+ 3 on CAFA6)	ProteInfer, DeepGOZero, DeepGOPlus, CLEAN, DeepProZyme, EnsBERT, METEOR	`pillar1_suite1_cafa_temporal.tsv`, `suite1_cafa6_prospective_summary.tsv`, `scenario1_*.tsv`
Suite 2	Pillars 1, 2, 4 / Scen. 2A–2C, C	9-gen. ablation (81 runs), 13/21 panels, 1,000 BeBoy, 88 4-king., 6 Euk.	3-ch (9/13/21/80) / 7-ch (21/1000/6 Euk.)	Stage 0 + 1 + 2A (A1–A9, E1–E8)	NCBI PGAP, InterProScan 5, metagenomic-deepFRI, DeepGO-SE, SPROF-GO, gLM2	`pillar1_2_ablation_summary.tsv`, `pillar1_suite2_*.tsv`, `pillar3_eukaryote_*.tsv`, `hyperparameter_*.tsv`
Suite 3	Pillars 2, 3 / Scen. 3A–3B (Str. A)	108 models, 6 BiGG GEMs, *E. coli* LRO (256/307), *B. subtilis* (144)	3-ch + Pfam + ESM-2 centroids	Stage 1 (RLGC/M4) + Stage 2B (SLR)	CarveMe, Reconstructor, METEOR v8, Positive-Cost LP, ESM-2 PLM Centroids	`pillar2_invariant_audit_summary.tsv`, `pillar2_curated6_gems_real.tsv`, `pillar2_lro15_summary.tsv`
Suite 4	Pillar 3 / Scen. 4 (Stratum B)	Role-stratified LRO Alternative Catabolic (*n* = 40)	3-ch + Pfam + ESM-2 centroids	Stage 1 (RLGC/M4) + Stage 2B (Cognate)	RLGC Kernel Density, Sequence-only ranking, METEOR v8	`pillar2_lro15_summary.tsv`, `pillar3_metabolism_lro.tsv`
Suite 5	Pillar 3 / Scen. 5 (Str. C), B. MAGs	LRO Secondary (*n* = 58), 106 BGCs, 175 LSO complexes (M1–M6), 97 MAGs	3-ch + Pfam + ESM-2 centroids	Stage 1 + Stage 2A + Stage 3 Agent	Baseline Noise-OR, Stage 1 Coherence/Operon, Minimal-Glucose MILP	`pillar2_3_antismash_106bgc_specificity.tsv`, `pillar3_complex_lso.tsv`, `pillar3_dark_matter_97mags.tsv`
Suite 6	Pillar 2 / Scen. 6 (Price-149)	148 host-mapped RB-TnSeq enzymes across 22 bacterial genomes (149 seqs)	1-pred & 5-pred ensemble	Single/Multi-pred + Stage 1 + Stage 2B	CLEAN, MAPred, DeepProZyme, EnsBERT, GraphEC, TopEC, METEOR, DCGM, ECrecer	`pillar2_price149_all5_real.tsv`, `pillar2_price149_by_stratum.tsv`
Suite 7	Pillar 3 / Scen. 7 (BacDive Trk 1)	529 carbon strains (2,322), 3,017 type strains (17,496), 6,313 strains (50,075)	3-ch + 3 GEM reconstructions	Stage 1+2 Multi-Model + DiASPora Hybrid	gapseq, ModelSEED, CarveMe, METEOR, Traitar, BacDive RF, DiASPora GBDT	`pillar3_bacdive_track1_summary.tsv`, `pillar3_bacdive_enzyme_per_ec.tsv`, `pillar3_bacdive_per_fold_cv.tsv`
Suite 8	Pillars 3, 5, 6 / Scen. 8 (Trk 2), Agent	Held-out 70% BacDive (1,641/12,316), 120 anomalies, 50 LRO, 5 isolates	3-ch (7-ch on Euk. anomalies)	Stage 2B Anchor + Stage 3 Agent	gapseq, Traitar, BacDive RF, DiASPora, Non-Agnostic LR/RF/GBDT/MLP Baselines	`pillar3_bacdive_track2_summary.tsv`, `pillar4_agent_*.tsv`, `pillar4_ml_baselines.tsv`

for the top $M_{\text{ess}} = 3$ candidates,

$$B_1(p,f) = \mathbb{I}[f \in \mathcal{F}_{\text{ess}}(\tau_{\text{tax}}) \wedge P_{\max}^{(k-1)}(f) < \theta_{\text{ess}}] \times \mathbb{I}[P^{(k-1)}(p,f) \geq 0.05 \wedge \text{rank}_f(p) \leq M_{\text{ess}}] \times 0.5^{\text{rank}_f(p)-1}\big(1 - P_{\max}^{(k-1)}(f)\big),$$

where the geometric rank decay $0.5^{\text{rank}_f(p)-1}$ in `EssentialityPrior.groovy` (or $1/\text{rank}_f(p)$ in the analytical Python ablation runner) concentrates the rescue boost on the top candidate.

2. **Pathway and complex coherence prior ($m = 2$, $\alpha_2 = 1.5$, $\beta_{\text{single}} = 1.2$, `CoherencePrior.groovy`):** For a pathway or complex $C \subseteq \mathcal{T}$ containing $f \in C$, let $c_{-f} = \text{cov}^{(k-1)}(C \setminus \{f\}) = \frac{1}{|C|-1}\sum_{f' \in C\setminus\{f\}} \mathbb{I}\big[\max_{p'} P^{(k-1)}(p', f') \geq 0.50\big]$. For $|C| \geq 3$, `CoherencePrior.groovy` boosts candidates when $c_{-f} \geq \theta_{\text{coh}} = 0.70$ with normalized ramp $(c_{-f} - \theta_{\text{coh}})/(1 - \theta_{\text{coh}})$ (plus $+0.25\, c_{-f}$ for 2-step modules when $c_{-f} \geq 0.50$) and penalizes isolated subunits in heteromeric complexes ($f \in \mathcal{C}_{\text{het}}$, $|C| \geq 3$) when $c_{-f} < \theta_{\text{het}} = 0.35$:

$$B_2(p,f) = \mathbb{I}\big[P^{(k-1)}(p,f) \in [0.05, 0.50)\big] \times \max_{C \ni f}\Big(\mathbb{I}[c_{-f} \geq \theta_{\text{coh}}]\frac{c_{-f} - \theta_{\text{coh}}}{1 - \theta_{\text{coh}}}\Big) - \frac{\beta_{\text{single}}}{\alpha_2}\mathbb{I}\big[P^{(k-1)}(p,f) \geq 0.05 \wedge f \in \mathcal{C}_{\text{het}}\big] \times \mathbb{I}[c_{-f} < \theta_{\text{het}}]\Big(1 - \frac{c_{-f}}{\theta_{\text{het}}}\Big).$$

3. **Taxonomic consistency prior ($m = 3$, $\alpha_3 = 3.0$, `ConsistencyPrior.groovy`):** Let $\mathcal{V}_{\text{SAT}}^{(k-1)}(p) \subseteq \mathcal{T}$ be the set of terms participating in minimal unsatisfiable subsets (MUS) extracted by SAT4J over the propositional encoding of $\mathcal{T}$, GO disjointness axioms, and direct GO taxon constraints for τ_{tax} on $\{f : P^{(k-1)}(p,f) \geq 0.20\}$; then $B_3(p,f) = -w_{\text{MUS}}(f)\,\mathbb{I}\big[f \in \mathcal{V}_{\text{SAT}}^{(k-1)}(p)\big]$ with $w_{\text{MUS}}(f) = 1.5$ for direct taxonomic violations and 1.0 for general disjointness conflicts.

4. **Anatomical prior ($m = 4$, $\alpha_4 = 4.0$, $\alpha_{\text{boost}} = 1.10$, `AnatomicalPrior.groovy`):** Let $\mathcal{T}_{\text{forb}}(\tau_{\text{tax}}) \subseteq \mathcal{T}$ be the set of GO terms anatomically forbidden in τ_{tax} (Section 4.4), and let $\mathcal{T}_{\text{act}}^{(k-1)}(\tau_{\text{tax}})$ be the set of developmental GO terms linked to an anatomical structure u that is taxonomically valid in τ_{tax} and supported by $n_{\text{sup}}(u) \geq N_{\text{struct}} = 3$ proteins with $P^{(k-1)}(p', f') \geq 0.60$; then with support ratio $\rho_{\text{sup}}(u) = \min(1.0, n_{\text{sup}}(u)/N_{\text{struct}})$,

$$B_4(p,f) = -\mathbb{I}\big[f \in \mathcal{T}_{\text{forb}}(\tau_{\text{tax}}) \wedge P^{(k-1)}(p,f) \geq 0.20\big] + \frac{\alpha_{\text{boost}}}{\alpha_4}\rho_{\text{sup}}(u)\mathbb{I}\big[f \in \mathcal{T}_{\text{act}}^{(k-1)}(\tau_{\text{tax}})\big] \times \mathbb{I}\big[P^{(k-1)}(p,f) \in [0.22, 0.49]\big].$$

5. **Metabolic gap-filling and RLGC prior ($m = 5$, $\alpha_5 = 1.2$, `GapFillingPrior.groovy` and `ReactionLocalContextSuggester.groovy`):** Given a missing pathway reaction r mapped to catalytic term f_r, the Reaction-Local Genomic Context (RLGC) score for candidate protein p with $P^{(k-1)}(p, f_r) \geq 0.01$ combines four bounded signals:

$$S_{\text{RLGC}}(p,r) = q_{\text{cat}}(p,r) + \lambda_{\text{BF}} \tanh\big(\tfrac{1}{2}\log_{10}\text{BF}_{\text{operon}}(p,r)\big) + \lambda_{\text{dom}} S_{\text{Pfam}}(p,r) + \lambda_{\text{PLM}} \max\big(0, S_{\text{ESM2}}(p,r)\big), \tag{3}$$

where $q_{\text{cat}}(p,r) \in (0,1)$ is the calibrated catalytic homology/HMM probability; $\text{BF}_{\text{operon}}(p,r)$ is the empirical Bayes factor (termed the *RLGC kernel density scorer* when evaluated standalone in Table 2) measuring Gaussian positional enrichment of genes annotated to pathway $W(r)$ within local neighborhood $\mathcal{N}_w(p)$ ($\pm w$ genes on the same contig, $w = 5$, kernel $K_{\text{pos}}(p,p') = \exp(-\Delta g(p,p')^2/8)$) relative to the genome-wide background prevalence $|W(r)|/N$ with smoothing $\varepsilon_{\text{bf}} = 0.05$ ($\lambda_{\text{BF}} = 0.35$; in `ReactionLocalContextSuggester.groovy`, the log-odds suggestion score weights the four standardized signals by $w_{\text{seq}} = 0.30, w_{\text{BF}} = 0.35, w_{\text{dom}} = 0.20, w_{\text{PLM}} = 0.15$):

$$\text{BF}_{\text{operon}}(p,r) = \frac{\varepsilon_{\text{bf}} + \sum_{p' \in \mathcal{N}_w^*(p)} K_{\text{pos}}(p,p')\,\mathbb{I}[p' \in W(r)]}{\varepsilon_{\text{bf}} + (|W(r)|/N)\sum_{p' \in \mathcal{N}_w^*(p)} K_{\text{pos}}(p,p')},$$

with $\mathcal{N}_w^*(p) = \mathcal{N}_w(p) \setminus \{p\}$; $S_{\text{Pfam}}(p,r) \in [0,1]$ scores EC top-level class compatibility from Pfam domains ($\lambda_{\text{dom}} = 0.20$); and $S_{\text{ESM2}}(p,r) \in [-1,1]$ is the cosine similarity between the ESM-2 embedding of protein p and the pre-t_0 Swiss-Prot reference

centroid of reaction r ($\lambda_{\mathrm{PLM}} = 0.45$). Ranking candidates for reaction r by descending $S_{\mathrm{RLGC}}(p, r)$ with rank $\mathrm{rank}_r(p) \in \{1, 2, \dots\}$ over the top $K_{\mathrm{gap}} = 3$ candidates, the prior shift is $B_5(p, f_r) = \mathbb{I}[\mathrm{rank}_r(p) \le K_{\mathrm{gap}}] \min\big(1.5, S_{\mathrm{RLGC}}(p, r)\big)/\mathrm{rank}_r(p)$ (with $+1.2\,(S_{\mathrm{RLGC}} - 0.5)$ bonus in `GapFillingPrior.groovy` when $S_{\mathrm{RLGC}} > 0.50$).

6. **Operon neighborhood prior ($m = 6$, $\alpha_6 = 1.0$, `GenomicContextPrior.groovy`):** Let $\mathcal{N}_{\mathrm{op}}(p)$ denote adjacent same-strand genes within intergenic distance $d_{p,p'} \le d_{\max} = 150$ bp[42], weighted by $w_{\mathrm{ig}}(d) = \exp(-d/75)$ when base-pair coordinates are supplied (or $1/(1 + \Delta\mathrm{idx})$ when only gene-index order is available; in `GenomicContextPrior.groovy`, normalized across same-operon pathway partners). When $P^{(k-1)}(p, f) \in [0.05, 0.50)$ for a term $f \in C$,

$$B_6(p, f) = \mathbb{I}\big[P^{(k-1)}(p, f) \in [0.05, 0.50)\big] \times \min\Big(1.0, \sum_{p' \in \mathcal{N}_{\mathrm{op}}(p)} \max_{f' \in C \setminus \{f\}} w_{\mathrm{ig}}(d_{p,p'}) \times P^{(k-1)}(p', f')\, \mathbb{I}\big[P^{(k-1)}(p', f') \ge 0.50\big]\Big).$$

Hyperparameter protocol and sensitivity analysis. All 28 pipeline constants across Stage 1 ($\alpha_1, \dots, \alpha_6$, β_{single}, α_{boost}, θ_{ess}, θ_{coh}, θ_{het}, M_{ess}, K_{gap}, λ_{BF}, λ_{dom}, λ_{PLM}, $d_{\max}$) and Stages 2A–2B (γ_{relax}, θ_{sub}, θ_{core}, α_{bio}, η_{null}, γ_{FBA}, γ_{damp}, $\beta_1, \dots, \beta_4$, including the individual $\beta_{1,2}$ and γ_{damp} rows) were fitted exclusively on the *E. coli* K-12 MG1655 development proteome via two-stage block coordinate search (first optimizing Stage 1 weights/thresholds over a 5-point coordinate grid maximizing 5-fold cross-validated LRO Hit@1 + GAEF consistency, then optimizing Stage 2A/2B stoichiometric parameters with Stage 1 frozen) and held strictly frozen across all non-*E. coli* benchmarks. Complete one-at-a-time sensitivity perturbations of ±20% across all 28 parameters on the 8 non-*E. coli* genomes of Table 1, the 144 *B. subtilis* LRO reactions, and the 22-genome `Price-149` panel (`results/hyperparameter_sensitivity_non_ecoli.tsv`) confirm that mean 8-genome micro $F_{\max}$ varies by ≤ 0.0008, GAEF by ≤ 0.0038, *B. subtilis* LRO Hit@1 by ≤ 1.32 percentage points, and `Price-149` CLEAN $F_{\max}$ by ≤ 0.0042.

4.4 Cross-ontology taxonomic and anatomical constraint bridge

To capture taxonomic restrictions entailed by multicellular eukaryotic developmental and physiological GO terms, we programmatically extracted five resource tables from the official GO, Uberon[63], Plant Ontology (PO)[66], Cell Ontology (CL)[67], and NCBI Taxonomy OBO/OWL releases via `ExtractAnatomicalConstraints.groovy`: (1) 26,911 direct GO taxon constraint rules (13,819 `only_in_taxon` and 13,092 `never_in_taxon`) across 18,159 distinct GO terms (`go-taxon-constraints.tsv`); (2) 2,715 equivalence and subclass cross-product axioms linking 2,674 distinct GO terms to 1,294 Uberon, PO, and CL classes across 30 object properties (`RO:0002296 results_in_development_of`, `RO:0002298 results_in_morphogenesis_of`, `RO:0002297 results_in_formation_of`, `RO:0002299 results_in_maturation_of`, `BFO:0000066 occurs_in`, and transport/organization relations; `go-anatomical-constraints.tsv`); (3) 8,331 taxon constraints (6,230 `only_in_taxon` and 2,101 `never_in_taxon`) across 1,247 Uberon, PO, and CL classes (`uberon-po-taxon-constraints.tsv`); and (4) 7,722 anatomical hierarchy edges across 2,855 Uberon/PO/CL classes (`uberon-po-hierarchy.tsv`) and 1,818 taxonomic hierarchy edges across 1,305 NCBI Taxonomy clades (`ncbi-taxon-hierarchy.tsv`). A manual audit of a stratified random sample of 100 extracted GO–Uberon/PO/CL taxon-entailment pairs against curated literature confirmed 99/100 (99.0%) agreement (with 1 ambiguous life-stage term).

4.5 Stage 2A: Deterministic symbolic enforcement and semantic ancestor relaxation

In `IntegrateCommand.groovy` and `ConsistencyEnforcer.groovy`, Stage 2A executes immediately after Stage 1 (and prior to Stage 2B) as a linear-time ($O(|\mathcal{P}| \cdot |\mathcal{T}|)$) deterministic graph-traversal pass over the Stage 1 probability matrix. When a predicted annotation (p, t) with $P^{(K)}(p, t) \ge \theta_{\mathrm{eval}} = 0.50$ (or unpenalized posterior $P_{\mathrm{unpen}}(p, t) \ge 0.50$ prior to soft constraint suppression) violates a direct GO taxon constraint or an entailed Uberon/PO/CL anatomical constraint in τ_{tax}, Stage 2A performs breadth-first search upward in $\mathcal{T}$ from t to identify the set of nearest valid ancestors $\mathcal{A}_{\mathrm{valid}}(t, \tau_{\mathrm{tax}}) \subseteq \mathrm{Anc}(t) \setminus \{\mathrm{GO{:}0008150}, \mathrm{GO{:}0003674}, \mathrm{GO{:}0005575}\}$ that satisfy all taxonomic and anatomical constraints for τ_{tax} (if multiple incomparable ancestors lie at the minimum shortest-path distance, the probability update is applied to each minimal valid ancestor $a \in \mathcal{A}_{\mathrm{valid}}(t, \tau_{\mathrm{tax}})$ and propagated upward; if $\mathcal{A}_{\mathrm{valid}}(t, \tau_{\mathrm{tax}}) = \emptyset$ because all non-root ancestors of t are themselves forbidden in τ_{tax}, semantic relaxation defaults to removal with $P_{\mathrm{Stage2A}}(p, t) = 0$):

$$P_{\mathrm{Stage2A}}(p, a) = \max\Big(P^{(K)}(p, a),\, \gamma_{\mathrm{relax}}\, P_{\mathrm{unpen}}(p, t)\Big), \qquad P_{\mathrm{Stage2A}}(p, t) = 0, \tag{4}$$

with $\gamma_{\mathrm{relax}} = 0.95$, after which P_{Stage2A} is converted back to log-odds $L_{\mathrm{Stage2A}}(p, f) = \log \frac{\max(\varepsilon, P_{\mathrm{Stage2A}}(p, f))}{1 - \min(1-\varepsilon, P_{\mathrm{Stage2A}}(p, f))}$. In addition, for any obligate heteromeric complex $C \in \mathcal{C}_{\mathrm{het}}$ whose subunit support—defined as the fraction of required component functions $f' \in C$ having

$\max_{p'} P_{\text{Stage2A}}(p', f') \geq 0.50$—remains below $\theta_{\text{het}} = 0.35$, isolated singleton predictions above 0.50 are deterministically attenuated to 0.45 and logged to an auditable action ledger.

4.6 Stage 2B: Flux-certified stoichiometric locus refinement

Stage 2B (`scripts/stoichiometric_locus_refiner.py`) operates over a sanitized universal metabolic network $\mathcal{U}_{\text{clean}}$ constructed from the ModelSEED biochemistry database (v2.6.1 cross-referenced to MetaNetX v4.4 and BiGG reaction/EC identifiers, flagging mass- or charge-imbalanced pool reactions with penalty $w_{\text{imb}}(j) = 25.0$). For every universal reaction $j \in \mathcal{U}_{\text{clean}}$, reference EC and KO annotations are compiled into a disjunctive normal form (DNF) of catalytic mechanisms $\mathcal{M}_{\text{dnf}}(j) = \{m_1, \ldots, m_{K_j}\}$, where each mechanism $m \in \mathcal{M}_{\text{dnf}}(j)$ is a conjunction of required catalytic roles or complex subunits $\mathcal{R}_{\text{sub}}(m) = \{u_1, \ldots, u_{D_m}\}$ ($D_m = 1$ for monomeric enzymes; $D_m \geq 2$ for multi-subunit complexes in Complex Portal/KEGG Modules such as urease `EC 3.5.1.5` $\alpha\beta\gamma$). First, the sparse role aggregator computes candidate locus support $s_{p,u} = P_{\text{Stage2A}}(p, u) \in (0, 1)$ for $p \in \mathcal{C}_{\text{cand}}(u)$ and aggregates across loci using Top-M_{iso} ($M_{\text{iso}} = 3$) Noisy-OR for isozyme disjunctions and a bottleneck-gated geometric mean for subunit conjunctions:

$$
\begin{aligned}
Q(u) &= 1 - \prod_{p \in \text{Top-}M_{\text{iso}}(\mathcal{C}_{\text{cand}}(u))} \big(1 - s_{p,u}\big), \\
S_{\text{mech}}(m) &= \begin{cases} \Big(\prod_{u \in \mathcal{R}_{\text{sub}}(m)} Q(u)\Big)^{1/D_m}, & \min_u Q(u) \geq \theta_{\text{sub}}, \\ \min_{u \in \mathcal{R}_{\text{sub}}(m)} Q(u), & \text{otherwise}, \end{cases} \\
S_{\text{rxn}}(j) &= \max_{m \in \mathcal{M}_{\text{dnf}}(j)} S_{\text{mech}}(m),
\end{aligned}
\tag{5}
$$

with subunit bottleneck gate $\theta_{\text{sub}} = 0.25$.

Second, reactions are partitioned into core draft reactions $\mathcal{R}_{\text{core}} = \{j : S_{\text{rxn}}(j) \geq \theta_{\text{core}} = 0.50\}$ and candidate repair reactions with strictly positive addition penalties $c_j^{\text{add}} = \lambda_{\text{add}}(1 - S_{\text{rxn}}(j)) + w_{\text{imb}}(j) + \epsilon_{\text{par}} > 0$ ($\lambda_{\text{add}} = 1.0$, $\epsilon_{\text{par}} = 10^{-3}$; because $w_{\text{imb}}(j) = 25.0 = 25\lambda_{\text{add}}$, mass-imbalanced reactions are effectively forbidden whenever any balanced repair path exists). Splitting reversible fluxes into non-negative forward and backward variables $v_j = v_j^+ - v_j^-$ ($v_j^+, v_j^- \geq 0$, $|v_j| = v_j^+ + v_j^-$) and computing the maximum achievable biomass flux $v_{\text{bio}}^{\max}(\mathcal{E}_{\text{med}})$ over the balanced universal network under medium $\mathcal{E}_{\text{med}}$, Stage 2B solves a two-step lexicographic pFBA linear program[73] in HiGHS (feasibility tolerance 10^{-9}): Step 1 minimizes $\sum_{j \notin \mathcal{R}_{\text{core}}} c_j^{\text{add}}(v_j^+ + v_j^-)$ subject to $\mathbf{Sv} = \mathbf{0}$ and $v_{\text{bio}} \geq \alpha_{\text{bio}}\, v_{\text{bio}}^{\max}(\mathcal{E}_{\text{med}})$ ($\alpha_{\text{bio}} = 0.10$) to obtain minimum repair cost $C^*(\mathcal{E}_{\text{med}})$; Step 2 minimizes total network flux $\sum_{j \in \mathcal{U}_{\text{clean}}}(v_j^+ + v_j^-)$ subject to $\sum_{j \notin \mathcal{R}_{\text{core}}} c_j^{\text{add}}(v_j^+ + v_j^-) \leq (1+\delta)C^*(\mathcal{E}_{\text{med}})$ ($\delta = 10^{-6}$) and $v_{\text{bio}} \geq \alpha_{\text{bio}}\, v_{\text{bio}}^{\max}(\mathcal{E}_{\text{med}})$. A non-core reaction $j \notin \mathcal{R}_{\text{core}}$ is certified as an active repair ($j \in \mathcal{J}_{\text{rep}}(\mathcal{E}_{\text{med}})$) if and only if $|v_j^{\text{pFBA}}(\mathcal{E}_{\text{med}})| > \epsilon_{\text{flux}} = 10^{-6}$ ($10^3\times$ above solver feasibility tolerance 10^{-9}; identical counts hold at 10^{-7}). Each repaired reaction $j \in \mathcal{J}_{\text{rep}}(\mathcal{E}_{\text{med}})$ receives a stoichiometric confidence weight $\pi_j^{\text{FBA}} = 0.5\,\mathbb{I}[j \in \mathcal{R}_{\text{FVA}}] + 0.3\,\mathbb{I}[j \in \mathcal{R}_{\text{KO}}] + 0.2\,\omega_{\text{ens}}(j) \in [0, 1]$, where $j \in \mathcal{R}_{\text{FVA}}$ if its minimum flux under flux variability analysis at 99% optimal repair cost exceeds ϵ_{flux}, $j \in \mathcal{R}_{\text{KO}}$ if constraining $v_j = 0$ reduces biomass below $\alpha_{\text{bio}} v_{\text{bio}}^{\max}$ without increasing repair cost by $> 10\%$, and $\omega_{\text{ens}}(j) = 1.0$ if j remains active under a 5% objective perturbation pass and 0.65 otherwise.

Third, the locus responsibility updater attributes certified reaction flux back to candidate gene loci against an explicit unassigned-gene (null) hypothesis $h_\varnothing$ with prior mass $\eta_{\text{null}} = 0.05$:

$$
\begin{aligned}
w_{j,u}(p) &= \left(\frac{s_{p,u}}{1 - s_{p,u}}\right)^{\beta_1} \text{BF}_{\text{operon}}(p, u)^{\beta_2} \\
&\quad \times \exp\big(\beta_3 C_{\text{syn}}(p, u) + \beta_4 S_{\text{ESM2}}(p, u)\big), \\
r_{j,u}(p) &= \frac{w_{j,u}(p)}{\eta_{\text{null}} + \sum_{p' \in \mathcal{C}_{\text{cand}}(u)} w_{j,u}(p')},
\end{aligned}
\tag{6}
$$

with $\beta_1 = 1.0, \beta_2 = 1.0, \beta_3 = 1.5, \beta_4 = 1.5$, where $C_{\text{syn}}(p, u) \in [0, 1]$ is the co-operonic pathway partner support. When three candidate genes carry only noise-floor sequence support ($s_{p,u} \leq 0.01, \text{BF}_{\text{operon}} = 1, C_{\text{syn}} = 0, S_{\text{ESM2}} = 0$), $\sum_{p'} w_{j,u}(p') \leq 0.0303 < \eta_{\text{null}} = 0.05$, so $h_\varnothing$ absorbs $r_{j,u}(h_\varnothing) \geq 0.62 > 0.50$ and prevents false-positive gene promotion; conversely, a single candidate with moderate homology ($s_{p,u} = 0.30$) already yields $w_{j,u}(p) \geq 0.429$, reducing $r_{j,u}(h_\varnothing)$ to ≤ 0.104 and attributing $r_{j,u}(p) \geq 0.896$ to the supported locus. In outer pass $r \in \{1, 2, 3\}$, candidate loci receive the stoichiometric log-odds shift $\Delta L_{\text{FBA}}^{(r)}(p, u) = \gamma_{\text{FBA}} \sum_{j \in \mathcal{J}_{\text{rep}}^{(r)}(u)} \pi_j^{\text{FBA}}\, r_{j,u}^{(r)}(p)\,(1 - S_{\text{rxn}}^{(r)}(j))$ ($\gamma_{\text{FBA}} = 2.2$). In **Full GSPA Iter** k ($k \in \{1, 2, 3\}$), outer pass k updates the Stage 1 anchor log-odds in Eq. (2) via geometrically damped accumulation ($\gamma_{\text{damp}} = 0.50$):

$$
L_{\text{anchor}}^{(k)}(p, u) = L^{(0)}(p, u) + \sum_{r=1}^{k-1} \gamma_{\text{damp}}^{r-1}\, \Delta L_{\text{FBA}}^{(r)}(p, u), \tag{7}
$$

and re-runs one pass of Stage 1 + Stage 2A + Stage 2B, contracting monotonically by outer Iteration 3.

To formalize the theoretical limits of metabolic gene recovery across Strata A, B, and C, we define five binary recoverability indicators for each held-out reaction j and true gene p^*: (1) $\mathcal{C}_{\text{struct}}(j) \in \{0, 1\}$, equal to 1 iff reaction j is present and mass-balanced in $\mathcal{U}_{\text{clean}}$; (2) $\mathcal{C}_{\text{task}}(j, \mathcal{E}_{\text{med}}) \in \{0, 1\}$, equal to 1 iff blocking j requires non-zero repair flux under medium $\mathcal{E}_{\text{med}}$ ($\mathcal{C}_{\text{task}} = 1$ on minimal glucose for Stratum A, $\mathcal{C}_{\text{task}} = 1$ on cognate substrate for Stratum B, and $\mathcal{C}_{\text{task}} = 0$ for secondary metabolism Stratum C);

(3) $\mathcal{C}_{\text{unique}}(j) \in \{0,1\}$, equal to 1 iff j has no unmasked isozyme sufficient to carry full flux; (4) $\mathcal{C}_{\text{operon}}(p^*, j) = \mathbb{I}[\text{BF}_{\text{operon}}(p^*, j) > 1]$; and (5) $\mathcal{C}_{\text{complex}}(p^*, j) \in \{0,1\}$, equal to 1 iff j is catalyzed by a multi-subunit complex with partner subunits present.

4.7 Stage 3: Selective neuro-symbolic agentic annotation completion

Stage 3 (`AnomalyExtractor.groovy` and `benchmark/agent/run_gspa_agent.py`) operates selectively on post-Stage-2 genome-scale anomalies across four categories: (1) `PATHWAY_GAP` (a pathway with completeness $\geq \theta_{\text{path_anom}}$ missing a step f with sub-threshold candidates in $P_{\text{Stage2}} \in [\theta_{\text{floor}}, 0.50)$, where `AnomalyExtractor.groovy` defaults to $\theta_{\text{path_anom}} = 0.50$, $\theta_{\text{floor}} = 0.05$, up to 8 candidates per anomaly and 150 anomalies per genome, and the Suite 8 benchmark focuses on high-completeness pathways ≥ 0.65 with top 5 candidates in $[0.15, 0.49]$); (2) `INCOMPLETE_COMPLEX` (an obligate complex with $\geq$ 50% subunit coverage missing a required subunit with borderline candidates); (3) `BORDERLINE_ESSENTIAL` (an unfulfilled core essential category $f \in \mathcal{F}_{\text{ess}}(\tau_{\text{tax}})$ with sub-threshold candidates); and (4) `BORDERLINE_CONFLICT` (a claim in $[0.35, 0.62]$ subject to a taxonomic or anatomical conflict where domain inspection can refine the ancestral target).

For each anomaly, Stage 3 runs a bounded multi-turn tool-calling loop ($\leq$ 6 turns in full mode, $\leq$ 2 turns in fast 1-step mode) using the open-weight instruction model `Qwen/Qwen3.8-27B-FP8`[84] (served as `qwen3.8-27b` via vLLM on `unimatrix01` with a 32,768-token context window, temperature 0.0, maximum 450 output tokens per turn; evaluated across 3 independent random seeds seed $\in \{42, 43, 44\}$, `results/pillar4_multiseed_summary.tsv`) equipped with five deterministic read-only tools: `inspect_operon_neighborhood`, `inspect_pathway_or_complex_gap`, `query_protein_evidence`, `check_symbolic_constraints`, and `propose_resolution`. Every proposed resolution is verified by the deterministic `SymbolicGuardrail` against all 26,911 direct GO taxon constraints and 2,715 GO–Uberon/PO/CL anatomical constraints before acceptance.

4.8 BacDive Track 1, Track 2 Bayesian Anchor, and GSPA + DiASPora Hybrid protocols

In Suite 7 (`Track 1, benchmark/unified/pillar3/run_bacdive_sidecar_benchmark.py`) for each strain s and target carbon substrate or diagnostic enzyme y, automated metabolic reconstructions (`CarveMe` v1.5.2[54], `ModelSEED` v2.6.1[52], and `gapseq` v1.2[55]) were executed as released to obtain binary pathway/reaction calls $m_k(s, y) \in \{0, 1\}$. **GSPA Track 1 (100% Blind)** takes `gapseq` as the base stoichiometric indicator ($P_{\text{base}}(s, y) = 0.75$ if $m_{\text{gapseq}}(s, y) = 1$ else 0.20), adjusts by +0.10 when `ModelSEED` and `CarveMe` both agree ($m_{\text{mseed}} = m_{\text{carveme}} = 1$) and −0.08 when both are absent ($m_{\text{mseed}} = m_{\text{carveme}} = 0$), and on carbon substrates adds +0.12 when ≥ 2 positive feeder assays support the strain's catabolic network (clamped to $[0.02, 0.98]$), using zero experimental phenotype training labels and preserving `gapseq`'s binary decision boundary at $\theta = 0.50$ while yielding a continuous confidence score for threshold-swept ROC-AUC evaluation. In Suite 8 (**GSPA Track 2 Bayesian Anchor**), 30% of each strain's experimental assays $\mathcal{Y}_{\text{rev}}(s)$ are revealed; let $\bar{p}_{\text{rev}}(s) = (|\mathcal{Y}_{\text{rev}}(s)| + 2)^{-1}\big(1 + \sum_{y' \in \mathcal{Y}_{\text{rev}}(s)} y'(s)\big)$ denote the Laplace-smoothed revealed strain activity rate, and let $r_{\text{pb}}(y', y)$ denote the point-biserial correlation between assay y' and assay y across non-test reference folds. The held-out 70% assays $y \in \mathcal{Y}_{\text{hold}}(s)$ are scored by combining $P_{\text{Track1}}(s, y)$ with the empirical Bayes strain prior and correlated revealed assays ($0.50\, P_{\text{Track1}} + 0.35\, \bar{p}_{\text{rev}} + 0.15\, \bar{c}_{\text{corr}}$ when $|\mathcal{Y}_{\text{rev}}(s)| \geq 2$).

For the supervised classifiers (`Traitar` L1-regularized linear SVM[81], `BacDive RF Baseline` Random Forest with 500 trees[79,80], and `DiASPora` Gradient-Boosted Decision Trees with 300 trees, max depth 6, learning rate 0.05[80]), all models were trained and evaluated under identical 5-fold genus-blocked cross-validation splits (where all strains of any NCBI genus are assigned to the same fold; per-fold metrics in `results/pillar3_bacdive_per_fold_cv.tsv`) over Pfam, KEGG Orthology (KO), and EC feature profiles. In **GSPA + DiASPora Hybrid**, out-of-genus probabilities from `DiASPora` (0.55 weight), `Traitar` (0.20), and `GSPA` (0.25, using P_{Track1} in Track 1 and P_{Track2} in Track 2) are combined strictly across out-of-fold predictions without test-genus leakage. All wall-clock runtimes and peak heap memory were measured on dual-socket AMD EPYC 7763 compute nodes (8 CPU threads per genome; NVIDIA A100-SXM4-80GB GPU for vLLM and PLM embedding extraction).

4.9 Evaluation metrics, donor-database homology masking, and internal GAEF score

Across Suites 1–8, functional predictions are evaluated using standard protein-centric CAFA metrics ($F_{\max}$, information-accretion-weighted $WF_{\max}$, pair-level micro $F_{\max}$, and minimum semantic distance $S_{\min}$ across thresholds $\theta \in \{0.01, \ldots, 0.99\}$[1,17,85]) with 95% bootstrap or exact Clopper–Pearson confidence intervals and two-sided paired Wilcoxon or McNemar tests. Under donor-database homology masking (`Protocol H` at sequence-identity cutoffs $\eta_{\text{id}} \in \{100\%, 70\%, 50\%, 30\%\}$ and taxonomic Genus/Family masking `Protocol D` in `apply_protocol_H.py`), all reference donor proteins r with pairwise sequence identity $\text{pident}(g, r) > \eta_{\text{id}}$ (or sharing the same genus/family; when explicit donor `pident`

metadata is absent on a pre-merged claim, raw homology confidence is conservatively capped at $\eta_{\text{id}}/100$) to any test protein g are excluded from the DIAMOND database, ESM-2/ProtT5 kNN donor indices, FoldSeek templates, and probability calibrators; for the neural enzyme predictors in Suite 1C (`CLEAN`, `DeepProZyme`, and `EnzBERT`), reference lookup databases and training-split homologs above η_{id} are filtered per each tool's homology-split evaluation protocol, while the underlying pretrained protein language model backbone weights (ESM-2 and ProtT5) remain frozen as released (which may retain pretraining exposure to UniRef50 sequences and therefore makes strict Track A cosine-masked LRO a more conservative test of de novo context recovery).

Complementing external ground-truth evaluation, the internal Genome-Scale Annotation Evaluation Framework (GAEF; `QualityReport.groovy`) structural audit score summarizes genome-wide invariants at $\theta_{\text{eval}} = 0.50$—Essential Function Completeness (C_{ess}), Functional Coherence (C_{func}, the mean of active Pathway Coherence C_{path}, Complex Coherence C_{cplx}, and Process Coherence C_{proc}, or $\frac{1}{2}(C_{\text{path}} + C_{\text{cplx}})$ when process coherence is omitted), and Taxonomic/Anatomical Consistency ($C_{\text{cons}} = \max(0,\, 1 - 0.01\, V(\mathcal{P}))$, where $V(\mathcal{P})$ is the count of formal constraint violations):

$$\text{GAEF}(\mathcal{P}) = 0.30\, C_{\text{ess}} + 0.40\, C_{\text{func}} + 0.30\, C_{\text{cons}}, \qquad (8)$$

which equals $0.30\, C_{\text{ess}} + 0.20\, C_{\text{path}} + 0.20\, C_{\text{cplx}} + 0.30\, C_{\text{cons}}$ on the two-component coherence panels. Because $C_{\text{ess}}, C_{\text{func}}$, and C_{cons} measure the structural invariants targeted by Stages 1 and 2, GAEF is reported strictly as an internal structural diagnostic alongside held-out experimental validation.

Data availability

All reference ontologies (Gene Ontology, Uberon, Plant Ontology, Cell Ontology, and NCBI Taxonomy), curated cross-ontology constraint tables (`go-taxon-constraints.tsv`, `go-anatomical-constraints.tsv`, `uberon-po-taxon-constraints.tsv`, `uberon-po-hierarchy.tsv`, `ncbi-taxon-hierarchy.tsv`, and `complex_terms.tsv`, derived from CC-BY-4.0 OBO Foundry ontologies and released under CC-BY-4.0), SHA-256-verified benchmark manifests (`benchmark/manifests/`), and frozen empirical result tables (`results/*.tsv` and `results/PROVENANCE_MAP.md`) are included in the repository at https://github.com/bio-ontology-research-group/gspa (release tag `v2.1.0`).

Code availability

The GSPA v2.1.0 source code, standalone command-line executable (`gspa-2.1.0.jar`), Stage 2B flux-certified stoichiometric locus refiner (`scripts/stoichiometric_locus_refiner.py`), Stage 3 neuro-symbolic agent and tool schemas (`benchmark/agent/run_gspa_agent.py`), figure generation scripts (`paper/generate_figures.py` and `paper/generate_figure6.py`), and benchmark reproduction scripts are openly available under the GPL-3.0-or-later license at https://github.com/bio-ontology-research-group/gspa (release tag `v2.1.0`).

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Author contributions

T.A.-S., M.K., and R.H. conceived and designed the study. T.A.-S. and R.H. implemented the GSPA v2.1.0 software, cross-ontology constraint bridge, stoichiometric locus refiner, and neuro-symbolic agent, executed the benchmarks, and analyzed the data. T.A.-S., M.K., and R.H. wrote and revised the manuscript.

Competing interests

The authors declare no competing interests.

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